

Software Requirements Specification (SRS)

ThinkPlus Adaptive Learning and Assessment Platform

Version 1.0 (Draft)

Prepared For

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Revision History

Name	Date	Reason For Changes	Version
Narendra Sirisipalli	12 JUNE 2026	Initial Update	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements for the ThinkPlus Adaptive Learning and Assessment Platform.

The purpose of this document is to provide a comprehensive description of the system requirements, constraints, interfaces, and expected behavior of the platform. It serves as the primary reference document for stakeholders, product owners, business analysts, software architects, developers, testers, project managers, and client representatives throughout the software development lifecycle.

The ThinkPlus Adaptive Learning and Assessment Platform is designed to provide personalized learning experiences through adaptive assessments, intelligent question selection, mastery tracking, ability estimation, performance analytics, and curriculum navigation. The platform shall continuously adapt learning paths based on student performance, mastery levels, and learning progression.

This document establishes a common understanding between ThinkPlus and the development team regarding the scope, objectives, requirements, and expected outcomes of the project. It also serves as the baseline for system design, implementation, testing, deployment, and maintenance activities.

The objectives of this SRS are to:

- Define the business and system requirements of the platform.
- Describe the functional capabilities expected from the system.
- Specify performance, security, reliability, and scalability requirements.
- Establish acceptance criteria for project deliverables.
- Provide traceability between business goals and implemented features.
- Reduce ambiguity and ensure consistency among all project stakeholders.

This SRS follows the principles and recommendations defined in ISO/IEC/IEEE 29148:2018 for requirements engineering and software specification practices.

1.2 Document Conventions

The following conventions are used throughout this document:

Requirement Statements

Mandatory requirements are expressed using the term:

"Shall"

Example:

"The system shall authenticate users before granting access to protected resources."

Recommended or desirable capabilities are expressed using:

"Should"

Example:

"The system should provide personalized learning recommendations."

Optional capabilities are expressed using:

"May"

Example:

"The system may support additional assessment formats in future releases."

Requirement Identification

Each requirement shall be assigned a unique identifier for traceability purposes.

Examples:

- FR-AUTH-001 (Functional Requirement)
- NFR-PERF-001 (Performance Requirement)
- SEC-001 (Security Requirement)
- ADP-001 (Adaptive Learning Requirement)

Terminology

Technical terms, abbreviations, and domain-specific terminology used throughout this document are defined in Appendix A: Glossary.

Priority Classification

Requirements may be categorized using the following priority levels:

Priority	Description
High	Critical for system operation and project success
Medium	Important but not mandatory for initial release
Low	Optional enhancement or future consideration

1.3 Intended Audience and Reading Suggestions

This document is intended for multiple stakeholder groups involved in the planning, development, deployment, and operation of the ThinkPlus Adaptive Learning and Assessment Platform.

Intended Audience

Stakeholder Group	Purpose
ThinkPlus Management	Business review and approval
Product Owners	Requirement validation and prioritization
Business Analysts	Requirement analysis and traceability
Solution Architects	System architecture design
Software Developers	System implementation
Content Management Teams	Question bank and curriculum preparation
Data Science Teams	Adaptive engine optimization and analytics
Client Representatives	Requirement review and acceptance

Reading Recommendations

Readers may focus on sections most relevant to their responsibilities:

- Management and Product Owners: Sections 1, 2, 3, and 17.
- Architects and Developers: Entire document.
- QA Teams: Sections 3, 4, 5, and 17.
- Content Teams: Sections 3.4, 3.5, 3.6, and 3.7.
- Data Science Teams: Sections 3.6 through 3.10 and Section 6.

1.4 Project Scope

The ThinkPlus Adaptive Learning and Assessment Platform is an AI-assisted educational platform designed to deliver personalized learning experiences through adaptive assessments, intelligent content sequencing, and data-driven learning analytics.

The platform shall support:

- Student learning and assessment activities.
- Adaptive question delivery.
- Mastery tracking at subtopic level.
- Ability estimation at topic level.
- Intelligent curriculum navigation.

- Performance analytics and reporting.
- Teacher and administrator management capabilities.
- Question bank and knowledge graph management.

The platform aims to improve learning outcomes by identifying individual strengths and weaknesses, adjusting content difficulty dynamically, and providing targeted learning recommendations.

The system shall incorporate adaptive learning methodologies, including:

- Bayesian Knowledge Tracing (BKT) for mastery estimation.
- Item Response Theory (IRT) for ability estimation.
- Expected A Posteriori (EAP) estimation techniques.
- Fisher Information-based question selection.
- Knowledge Graph-based navigation and prerequisite management.
- Intelligent backtracking and remediation mechanisms.

The platform is intended for educational institutions, training organizations, aptitude preparation programs, competitive examination preparation, and skill development initiatives.

1.5 References

The following standards, publications, and resources are referenced in this document:

1. ISO/IEC/IEEE 29148:2018 – Systems and Software Engineering – Life Cycle Processes – Requirements Engineering.
2. ISO/IEC 25010:2011 – Systems and Software Quality Requirements and Evaluation (SQuaRE).
3. Corbett, A. T., & Anderson, J. R. (1994). Knowledge Tracing: Modeling the Acquisition of Procedural Knowledge.
4. Lord, F. M. (1980). Applications of Item Response Theory to Practical Testing Problems.
5. Bock, R. D., & Mislevy, R. J. (1982). Adaptive EAP Estimation of Ability in a Content-Structured Test.
6. OWASP Application Security Verification Standard (ASVS).
7. REST Architectural Style Guidelines.
8. PostgreSQL Documentation.
9. FastAPI Framework Documentation.
10. Next.js Framework Documentation.

2. Overall Description

2.1 Product Perspective

The ThinkPlus Adaptive Learning and Assessment Platform is a web-based educational technology solution designed to deliver personalized learning experiences through adaptive assessments, intelligent content recommendation, mastery tracking, and performance analytics.

The platform serves as a centralized learning ecosystem that enables students to practice and improve their knowledge through dynamically adjusted learning paths. The system continuously evaluates student performance and adapts content delivery based on learning progress, mastery levels, and estimated ability.

The platform consists of the following major subsystems:

- Authentication and Authorization System
- Student Learning Portal
- Teacher Portal
- Administrator Portal
- Question Bank Management System
- Topic and Knowledge Graph Management System
- Adaptive Learning Engine
- Bayesian Knowledge Tracing (BKT) Engine
- IRT and EAP Ability Estimation Engine
- Fisher Information Question Selection Engine
- Navigation and Backtracking Engine
- Analytics and Reporting Engine

The system shall operate as a cloud-based platform accessible through modern web browsers and shall support scalable deployment for educational institutions, training organizations, and assessment providers.

2.2 Product Features

The ThinkPlus Adaptive Learning and Assessment Platform provides the following high-level features:

User Management

- User registration and authentication.
- Role-based access control.
- Student, Teacher, and Administrator profiles.

Learning Management

- Personalized learning sessions.
- Adaptive assessment delivery.
- Learning progress tracking.
- Topic and subtopic mastery monitoring.

Question Bank Management

- Question creation and maintenance.
- Topic and subtopic mapping.
- Difficulty level classification.
- Question review and approval workflows.

Adaptive Learning Engine

- Ability estimation using Item Response Theory (IRT).
- Expected A Posteriori (EAP) ability updates.
- Fisher Information-based question selection.
- Adaptive difficulty adjustment.

Knowledge Tracking

- Bayesian Knowledge Tracing (BKT) for mastery estimation.
- Subtopic mastery monitoring.
- Learning progression analysis.

Navigation and Remediation

- Knowledge graph-based navigation.
- Prerequisite topic management.
- Intelligent backtracking and remediation.

Analytics and Reporting

- Student performance dashboards.
- Topic-level analytics.
- Mastery reports.
- Administrative reports and insights.

2.3 User Classes and Characteristics

The system shall support the following user classes:

Student

Students are the primary users of the platform.

Characteristics:

- Participate in assessments and learning activities.
- Receive adaptive question recommendations.
- View learning progress and performance analytics.
- Follow personalized learning paths.

Technical Expertise:

- Basic computer and internet usage skills.

Teacher

Teachers are responsible for monitoring and supporting student learning.

Characteristics:

- Review student performance.
- Monitor topic mastery and progress.
- Analyze learning outcomes.
- Provide instructional support.

Technical Expertise:

- Intermediate computer proficiency.

Administrator

Administrators manage platform operations and system configuration.

Characteristics:

- Manage users and permissions.
- Configure topics and learning structures.
- Manage question banks.
- Access platform-wide reports and analytics.

Technical Expertise:

- Advanced system administration knowledge.

Content Manager

Content Managers maintain educational content and assessment materials.

Characteristics:

- Create and update questions.
- Manage topic and subtopic structures.
- Review content quality.
- Assign difficulty levels.

Technical Expertise:

- Educational content development experience.

Management Users

Management users require strategic reporting and analytics.

Characteristics:

- Review organizational performance metrics.
- Monitor platform usage statistics.
- Evaluate learning effectiveness.

Technical Expertise:

- Basic platform usage knowledge.

2.4 Operating Environment

The ThinkPlus Adaptive Learning and Assessment Platform shall operate within the following environment:

Client Environment

- Modern web browsers.
- Desktop and laptop computers.
- Tablets and mobile devices.
- Internet connectivity required.

Supported Browsers:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

Application Environment

Frontend Technology:

- Next.js
- TypeScript

Backend Technology:

- Python
- FastAPI

Validation Framework:

- Pydantic

Authentication and Security:

- JWT Authentication
- bcrypt Password Hashing

Database:

- PostgreSQL

Deployment Environment:

- Cloud-hosted infrastructure
- Linux-based servers
- Containerized deployment support

2.5 Design and Implementation Constraints

The system shall comply with the following constraints:

Technology Constraints

Frontend:

- Next.js framework.
- TypeScript language.

Backend:

- Python programming language.
- FastAPI framework.

Database:

- PostgreSQL relational database.

Security:

- JWT-based authentication.
- bcrypt password hashing.

Architectural Constraints

- Modular service-oriented architecture.
- Separation of adaptive learning logic from presentation layer.
- Scalable and maintainable code structure.
- RESTful API communication.

Regulatory Constraints

- Secure storage of user information.
- Compliance with applicable educational data protection policies.
- Protection of assessment integrity.

2.6 User Documentation

The system shall provide the following user documentation:

Student Documentation

- User Guide.
- Assessment Participation Guide.
- Learning Progress Guide.

Teacher Documentation

- Teacher User Manual.
- Student Monitoring Guide.
- Reporting Guide.

Administrator Documentation

- Administrator Manual.
- System Configuration Guide.
- User Management Guide.

Technical Documentation

- API Documentation.
- Deployment Documentation.
- System Administration Guide.

2.7 Assumptions and Dependencies

The following assumptions and dependencies apply to the project:

Assumptions

- Users have access to internet-enabled devices.
- Questions are correctly mapped to topics and subtopics.
- Educational content is validated by subject matter experts.
- Users possess basic digital literacy skills.
- Required cloud infrastructure is available.

Dependencies

- PostgreSQL database availability.
- FastAPI backend services.
- Authentication services.
- Knowledge graph configuration.
- Question bank availability.
- Cloud hosting infrastructure.
- Reporting and analytics services.

The successful operation of the adaptive learning engine depends on the availability of valid question metadata, topic structures, and learning content definitions.

3.1 Authentication & Authorization Module

3.1.1 Description and Priority

Description

The Authentication and Authorization Module is responsible for managing user identity verification, account access, session management, and role-based access control within the ThinkPlus Adaptive Learning and Assessment Platform.

This module ensures that only authenticated and authorized users can access protected resources and platform functionality. The module supports secure login, logout, password management, account recovery, session validation, and user role enforcement.

The system shall support multiple user roles including:

- Student
- Teacher
- Content Manager
- Administrator
- Management User

The module shall implement Role-Based Access Control (RBAC) to ensure users can only access resources and functions permitted by their assigned roles.

Authentication shall be performed using secure credentials and token-based session management mechanisms.

Priority

Priority Level: High

Rationale

Authentication and authorization are critical system components required to protect user information, learning data, assessment content, and administrative functionality. The platform cannot operate securely without this module.

3.1.2 Stimulus / Response Sequences

Use Case 1: User Login

Stimulus

A registered user enters valid login credentials and submits a login request.

Response

The system shall:

1. Validate user credentials.
2. Verify account status.
3. Generate a secure authentication token.
4. Establish a user session.
5. Load user role permissions.
6. Redirect the user to the appropriate dashboard.

Use Case 2: Invalid Login Attempt

Stimulus

A user enters invalid credentials.

Response

The system shall:

1. Reject the authentication request.
2. Display an appropriate error message.

3. Log the failed login attempt.
4. Increment failed login counters.
5. Apply security policies when repeated failures occur.

Use Case 3: Password Reset Request

Stimulus

A user requests password recovery.

Response

The system shall:

1. Verify the user's identity.
2. Generate a secure password reset token.
3. Send password reset instructions.
4. Allow secure password update.
5. Invalidate the reset token after successful completion.

Use Case 4: Unauthorized Resource Access

Stimulus

A user attempts to access a resource without sufficient permissions.

Response

The system shall:

1. Validate user authorization.
2. Deny access to the requested resource.
3. Record the unauthorized access attempt.
4. Display an appropriate access denied message.

Use Case 5: User Logout

Stimulus

A user initiates logout.

Response

The system shall:

1. Terminate the active session.
2. Invalidate authentication tokens.
3. Clear active session information.
4. Redirect the user to the login page.

3.1.3 Functional Requirements

User Registration

FR-AUTH-001

The system shall allow authorized users to create new user accounts.

FR-AUTH-002

The system shall store user account information securely.

FR-AUTH-003

The system shall enforce unique usernames and email addresses.

User Authentication

FR-AUTH-004

The system shall authenticate users using valid credentials.

FR-AUTH-005

The system shall validate user credentials before granting access.

FR-AUTH-006

The system shall deny access when credentials are invalid.

FR-AUTH-007

The system shall support secure session creation after successful authentication.

Session Management

FR-AUTH-008

The system shall create a unique session for each authenticated user.

FR-AUTH-009

The system shall terminate inactive sessions after a configurable timeout period.

FR-AUTH-010

The system shall invalidate sessions after logout.

FR-AUTH-011

The system shall prevent simultaneous unauthorized session reuse.

Password Management

FR-AUTH-012

The system shall support password reset functionality.

FR-AUTH-013

The system shall enforce configurable password complexity requirements.

FR-AUTH-014

The system shall securely store passwords using industry-standard hashing mechanisms.

FR-AUTH-015

The system shall allow users to change their passwords after authentication.

Authorization and Access Control

FR-AUTH-016

The system shall implement Role-Based Access Control (RBAC).

FR-AUTH-017

The system shall restrict access based on assigned user roles.

FR-AUTH-018

The system shall validate permissions before executing protected actions.

FR-AUTH-019

The system shall prevent unauthorized access to administrative functions.

FR-AUTH-020

The system shall support configurable permission assignments for administrative users.

Security Logging

FR-AUTH-021

The system shall record successful login events.

FR-AUTH-022

The system shall record failed login attempts.

FR-AUTH-023

The system shall record password reset activities.

FR-AUTH-024

The system shall record unauthorized access attempts.

FR-AUTH-025

The system shall maintain audit logs for authentication-related activities.

Account Management**FR-AUTH-026**

The system shall allow administrators to activate user accounts.

FR-AUTH-027

The system shall allow administrators to deactivate user accounts.

FR-AUTH-028

The system shall allow administrators to update user roles.

FR-AUTH-029

The system shall allow administrators to view user account status.

FR-AUTH-030

The system shall maintain user profile information associated with each account.

3.2 Student Learning Module

3.2.1 Description and Priority

Description

The Student Learning Module is the primary user-facing component of the ThinkPlus Adaptive Learning and Assessment Platform. This module enables students to participate in adaptive learning sessions, practice assessments, monitor learning progress, view performance analytics, and receive personalized learning recommendations.

The module acts as the interaction layer between students and the adaptive learning engine. It continuously tracks learning activities and provides individualized learning experiences based on mastery levels, estimated ability, and curriculum progression.

The Student Learning Module shall support:

- Student dashboard access.
- Learning session initiation.
- Topic and subtopic progression.
- Adaptive assessment participation.
- Learning history tracking.
- Progress monitoring.
- Performance visualization.
- Personalized recommendations.
- Achievement tracking.
- Learning goal management.

Priority

Priority Level: High

Rationale

The Student Learning Module represents the core business functionality of the platform. It directly impacts student engagement, learning effectiveness, and overall platform success.

3.2.2 Stimulus / Response Sequences

Use Case 1: Student Starts Learning Session

Stimulus

A student selects a topic or learning path and initiates a learning session.

Response

The system shall:

1. Verify student authentication.
2. Retrieve current learning state.
3. Retrieve mastery information.
4. Retrieve ability estimates.
5. Determine the appropriate starting point.
6. Generate a personalized learning session.
7. Present the first question or learning activity.

Use Case 2: Student Answers a Question

Stimulus

A student submits an answer to a question.

Response

The system shall:

1. Validate the response.
2. Evaluate answer correctness.
3. Record the interaction.
4. Update learning progress.
5. Update performance statistics.
6. Trigger adaptive learning processes.
7. Determine the next activity.
8. Display feedback to the student.

Use Case 3: Student Views Progress Dashboard

Stimulus

A student requests progress information.

Response

The system shall:

1. Retrieve learning statistics.
2. Retrieve mastery information.
3. Retrieve topic progress.
4. Retrieve assessment history.
5. Generate progress visualizations.
6. Display performance insights.

Use Case 4: Student Completes Learning Session

Stimulus

A student completes a learning session.

Response

The system shall:

1. Save session results.
2. Update progress metrics.
3. Generate performance summaries.
4. Recommend future learning activities.
5. Display session completion information.

Use Case 5: Student Resumes Learning

Stimulus

A student returns after a previous session.

Response

The system shall:

1. Retrieve saved progress.
2. Restore learning state.
3. Identify pending activities.
4. Continue from the appropriate learning position.

3.2.3 Functional Requirements

Student Dashboard

FR-STU-001

The system shall provide a personalized student dashboard.

FR-STU-002

The dashboard shall display current learning progress.

FR-STU-003

The dashboard shall display completed topics and subtopics.

FR-STU-004

The dashboard shall display ongoing learning activities.

FR-STU-005

The dashboard shall display recent assessment results.

Learning Session Management

FR-STU-006

The system shall allow students to start learning sessions.

FR-STU-007

The system shall maintain session continuity.

FR-STU-008

The system shall save learning progress automatically.

FR-STU-009

The system shall allow interrupted sessions to be resumed.

FR-STU-010

The system shall track session duration.

Topic Progression**FR-STU-011**

The system shall track student progress at topic level.

FR-STU-012

The system shall track student progress at subtopic level.

FR-STU-013

The system shall display completed topics.

FR-STU-014

The system shall display active learning topics.

FR-STU-015

The system shall display recommended next topics.

Learning Activities**FR-STU-016**

The system shall present questions during learning sessions.

FR-STU-017

The system shall collect student responses.

FR-STU-018

The system shall validate submitted responses.

FR-STU-019

The system shall provide feedback after answer submission.

FR-STU-020

The system shall support multiple question formats.

Progress Tracking

FR-STU-021

The system shall track topic completion percentages.

FR-STU-022

The system shall track subtopic completion percentages.

FR-STU-023

The system shall maintain historical learning records.

FR-STU-024

The system shall track assessment participation history.

FR-STU-025

The system shall track learning activity history.

Performance Monitoring

FR-STU-026

The system shall calculate student performance metrics.

FR-STU-027

The system shall maintain historical performance records.

FR-STU-028

The system shall display performance trends.

FR-STU-029

The system shall display strengths and weaknesses.

FR-STU-030

The system shall display learning recommendations.

Personalized Learning

FR-STU-031

The system shall provide personalized learning paths.

FR-STU-032

The system shall recommend learning activities based on student performance.

FR-STU-033

The system shall recommend remediation activities when required.

FR-STU-034

The system shall recommend advancement activities for high-performing students.

FR-STU-035

The system shall adapt learning experiences based on student interactions.

Learning History

FR-STU-036

The system shall maintain a complete learning history for each student.

FR-STU-037

The system shall allow students to review completed activities.

FR-STU-038

The system shall store assessment results.

FR-STU-039

The system shall store session summaries.

FR-STU-040

The system shall maintain progress records across multiple sessions.

Notifications and Recommendations

FR-STU-041

The system shall notify students about recommended learning activities.

FR-STU-042

The system shall notify students about incomplete learning paths.

FR-STU-043

The system shall notify students when learning milestones are achieved.

FR-STU-044

The system shall display personalized recommendations.

FR-STU-045

The system shall support future notification enhancements.

Reporting and Insights**FR-STU-046**

The system shall provide topic-level progress reports.

FR-STU-047

The system shall provide subtopic-level progress reports.

FR-STU-048

The system shall provide assessment history reports.

FR-STU-049

The system shall provide performance summaries.

FR-STU-050

The system shall provide learning insights to support student improvement.

3.3 Assessment Management Module

3.3.1 Description and Priority

Description

The Assessment Management Module is responsible for managing the complete lifecycle of assessments within the ThinkPlus Adaptive Learning and Assessment Platform. This module enables the creation, scheduling, delivery, monitoring, evaluation, and reporting of assessments.

The module shall support both traditional assessments and adaptive assessments. Assessments may be configured for practice, diagnostics, topic evaluation, subtopic evaluation, placement testing, and learning progression monitoring.

The Assessment Management Module acts as the orchestration layer between students, question banks, adaptive learning components, and reporting systems.

The module shall support:

- Assessment creation and configuration.
- Assessment scheduling.
- Assessment delivery.
- Adaptive assessments.
- Practice assessments.
- Topic-wise assessments.
- Diagnostic assessments.
- Timed assessments.
- Assessment submission and evaluation.
- Assessment result generation.
- Assessment history management.

Priority

Priority Level: High

Rationale

Assessments form the primary mechanism through which student knowledge, mastery, and ability are measured. Accurate assessment management is critical for adaptive learning effectiveness and learning analytics.

3.3.2 Stimulus / Response Sequences

Use Case 1: Student Starts Assessment

Stimulus

A student selects an assessment and initiates the assessment session.

Response

The system shall:

1. Verify student authentication.
2. Validate assessment eligibility.
3. Retrieve assessment configuration.
4. Generate assessment session.
5. Initialize timing controls if applicable.
6. Present the first assessment question.

Use Case 2: Student Answers Question

Stimulus

A student submits a response to an assessment question.

Response

The system shall:

1. Validate the response.
2. Record the answer.
3. Save interaction data.
4. Update assessment progress.
5. Determine the next question.
6. Present the next question.

Use Case 3: Assessment Time Expires

Stimulus

The configured assessment time limit is reached.

Response

The system shall:

1. Automatically terminate the assessment.
2. Save all submitted responses.
3. Mark unanswered questions appropriately.

4. Generate assessment results.
5. Display completion information.

Use Case 4: Student Submits Assessment

Stimulus

A student manually submits an assessment.

Response

The system shall:

1. Validate submission request.
2. Finalize assessment session.
3. Evaluate responses.
4. Calculate assessment scores.
5. Generate assessment results.
6. Store assessment records.

Use Case 5: Teacher Creates Assessment

Stimulus

A teacher or administrator creates a new assessment.

Response

The system shall:

1. Validate assessment configuration.
2. Validate question selection criteria.
3. Store assessment metadata.
4. Publish the assessment for student use.

3.3.3 Functional Requirements

Assessment Creation

FR-ASM-001

The system shall allow authorized users to create assessments.

FR-ASM-002

The system shall allow assessments to be associated with topics and subtopics.

FR-ASM-003

The system shall allow assessment descriptions to be defined.

FR-ASM-004

The system shall allow assessment objectives to be specified.

FR-ASM-005

The system shall assign unique identifiers to assessments.

Assessment Configuration

FR-ASM-006

The system shall allow assessment duration to be configured.

FR-ASM-007

The system shall allow question count configuration.

FR-ASM-008

The system shall support difficulty-level configuration.

FR-ASM-009

The system shall support assessment availability periods.

FR-ASM-010

The system shall support assessment status management.

Assessment Types

FR-ASM-011

The system shall support practice assessments.

FR-ASM-012

The system shall support diagnostic assessments.

FR-ASM-013

The system shall support topic-level assessments.

FR-ASM-014

The system shall support subtopic-level assessments.

FR-ASM-015

The system shall support adaptive assessments.

Assessment Delivery

FR-ASM-016

The system shall deliver questions to students during assessments.

FR-ASM-017

The system shall maintain assessment state during active sessions.

FR-ASM-018

The system shall support sequential question delivery.

FR-ASM-019

The system shall support adaptive question delivery when enabled.

FR-ASM-020

The system shall track assessment progress.

Timing and Monitoring

FR-ASM-021

The system shall support timed assessments.

FR-ASM-022

The system shall display remaining assessment time.

FR-ASM-023

The system shall automatically submit assessments upon time expiration.

FR-ASM-024

The system shall record assessment start times.

FR-ASM-025

The system shall record assessment completion times.

Response Management

FR-ASM-026

The system shall record student responses.

FR-ASM-027

The system shall maintain answer histories.

FR-ASM-028

The system shall support answer updates before submission.

FR-ASM-029

The system shall validate response formats.

FR-ASM-030

The system shall securely store assessment responses.

Assessment Evaluation

FR-ASM-031

The system shall evaluate submitted responses.

FR-ASM-032

The system shall determine correctness of responses.

FR-ASM-033

The system shall calculate assessment scores.

FR-ASM-034

The system shall calculate topic-level performance metrics.

FR-ASM-035

The system shall calculate subtopic-level performance metrics.

Assessment Results

FR-ASM-036

The system shall generate assessment result summaries.

FR-ASM-037

The system shall display student scores.

FR-ASM-038

The system shall display topic-wise performance.

FR-ASM-039

The system shall display subtopic-wise performance.

FR-ASM-040

The system shall store assessment results for future reference.

Assessment History**FR-ASM-041**

The system shall maintain assessment history records.

FR-ASM-042

The system shall allow students to review completed assessments.

FR-ASM-043

The system shall allow teachers to review assessment results.

FR-ASM-044

The system shall maintain assessment audit records.

FR-ASM-045

The system shall support historical performance analysis.

Adaptive Assessment Support**FR-ASM-046**

The system shall support integration with the Adaptive Learning Engine.

FR-ASM-047

The system shall support integration with the BKT Engine.

FR-ASM-048

The system shall support integration with the IRT & EAP Engine.

FR-ASM-049

The system shall support integration with the Fisher Information Question Selection Engine.

FR-ASM-050

The system shall provide assessment data to analytics and reporting services.

3.4 Question Bank Management Module

3.4.1 Description and Priority

Description

The Question Bank Management Module serves as the centralized repository for all assessment and learning content used within the ThinkPlus Adaptive Learning and Assessment Platform.

This module is responsible for creating, storing, organizing, validating, maintaining, and managing questions and their associated metadata. The module ensures that all questions are properly categorized and available for adaptive learning, assessments, analytics, and reporting.

The Question Bank shall act as the primary content source for:

- Learning Sessions
- Practice Assessments
- Diagnostic Assessments
- Adaptive Assessments
- Topic Evaluations
- Subtopic Evaluations
- Mastery Tracking Activities

Each question shall be associated with appropriate metadata to support adaptive learning and content management processes.

The module shall support:

- Question creation and editing.
- Question categorization.
- Topic and subtopic mapping.
- Difficulty level assignment.
- Question approval workflows.
- Bulk question imports.
- Question version control.
- Question status management.
- Question search and filtering.
- Question auditing and reporting.

Priority

Priority Level: High

Rationale

The quality and organization of the question bank directly affect learning effectiveness, assessment quality, adaptive engine performance, and reporting accuracy. This module forms the foundation of the adaptive learning ecosystem.

3.4.2 Stimulus / Response Sequences

Use Case 1: Create Question

Stimulus

A content manager creates a new question.

Response

The system shall:

1. Validate question information.
2. Validate topic and subtopic mappings.
3. Validate answer keys.
4. Store question metadata.
5. Assign a unique question identifier.
6. Save the question in draft status.

Use Case 2: Update Question

Stimulus

A content manager edits an existing question.

Response

The system shall:

1. Validate updated content.
2. Maintain version history.
3. Record modification information.
4. Save the updated version.

Use Case 3: Approve Question

Stimulus

An authorized reviewer approves a question.

Response

The system shall:

1. Validate approval permissions.
2. Change question status to approved.
3. Make the question available for assessments and learning activities.
4. Record approval audit information.

Use Case 4: Import Questions

Stimulus

A content manager uploads a question dataset.

Response

The system shall:

1. Validate file structure.
2. Validate question records.
3. Detect duplicate questions.
4. Import valid questions.
5. Generate import summary reports.

Use Case 5: Retrieve Questions

Stimulus

An assessment or adaptive engine requests questions.

Response

The system shall:

1. Apply filtering criteria.
2. Retrieve matching questions.
3. Return question metadata.
4. Log retrieval activities.

3.4.3 Functional Requirements

Question Creation

FR-QBM-001

The system shall allow authorized users to create questions.

FR-QBM-002

The system shall assign a unique identifier to each question.

FR-QBM-003

The system shall support question creation through web-based interfaces.

FR-QBM-004

The system shall validate mandatory question attributes before saving.

FR-QBM-005

The system shall support draft question creation.

Question Structure

FR-QBM-006

The system shall store question text.

FR-QBM-007

The system shall store answer options.

FR-QBM-008

The system shall store correct answer keys.

FR-QBM-009

The system shall support explanations and solution content.

FR-QBM-010

The system shall support question metadata storage.

Topic and Subtopic Mapping

FR-QBM-011

The system shall associate questions with topics.

FR-QBM-012

The system shall associate questions with subtopics.

FR-QBM-013

The system shall validate topic assignments.

FR-QBM-014

The system shall validate subtopic assignments.

FR-QBM-015

The system shall support future multi-skill question mapping.

Difficulty Management**FR-QBM-016**

The system shall support difficulty classifications.

FR-QBM-017

The system shall support Easy, Medium, and Hard difficulty levels.

FR-QBM-018

The system shall allow administrators to modify difficulty classifications.

FR-QBM-019

The system shall store expert-defined difficulty values.

FR-QBM-020

The system shall support future empirical difficulty recalibration.

Question Status Management**FR-QBM-021**

The system shall support Draft status.

FR-QBM-022

The system shall support Review status.

FR-QBM-023

The system shall support Approved status.

FR-QBM-024

The system shall support Archived status.

FR-QBM-025

The system shall prevent unapproved questions from being used in assessments.

Question Approval Workflow**FR-QBM-026**

The system shall support question review processes.

FR-QBM-027

The system shall allow reviewers to approve questions.

FR-QBM-028

The system shall allow reviewers to reject questions.

FR-QBM-029

The system shall maintain review history.

FR-QBM-030

The system shall record approval audit logs.

Question Version Management**FR-QBM-031**

The system shall maintain question version history.

FR-QBM-032

The system shall store modification timestamps.

FR-QBM-033

The system shall store modification authors.

FR-QBM-034

The system shall support rollback to previous versions.

FR-QBM-035

The system shall preserve historical question records.

Search and Retrieval

FR-QBM-036

The system shall support question searching.

FR-QBM-037

The system shall support filtering by topic.

FR-QBM-038

The system shall support filtering by subtopic.

FR-QBM-039

The system shall support filtering by difficulty level.

FR-QBM-040

The system shall support filtering by approval status.

Bulk Import and Export

FR-QBM-041

The system shall support bulk question import.

FR-QBM-042

The system shall support spreadsheet-based imports.

FR-QBM-043

The system shall validate imported records.

FR-QBM-044

The system shall generate import validation reports.

FR-QBM-045

The system shall support question export functionality.

Question Metadata

FR-QBM-046

The system shall maintain metadata for each question.

FR-QBM-047

The system shall store topic information.

FR-QBM-048

The system shall store subtopic information.

FR-QBM-049

The system shall store difficulty classifications.

FR-QBM-050

The system shall store creation and modification audit information.

Adaptive Learning Support**FR-QBM-051**

The system shall provide question metadata to the Adaptive Learning Engine.

FR-QBM-052

The system shall provide topic information to the BKT Engine.

FR-QBM-053

The system shall provide difficulty information to the IRT Engine.

FR-QBM-054

The system shall provide eligible question pools to the Fisher Information Question Selection Engine.

FR-QBM-055

The system shall support future psychometric parameter storage including discrimination, difficulty, and guessing parameters.

Audit and Compliance**FR-QBM-056**

The system shall maintain complete audit records for all question activities.

FR-QBM-057

The system shall record question creation events.

FR-QBM-058

The system shall record modification events.

FR-QBM-059

The system shall record approval events.

FR-QBM-060

The system shall maintain historical activity logs for compliance and reporting purposes.

3.5 Topic & Knowledge Graph Management Module

3.5.1 Description and Priority

Description

The Topic and Knowledge Graph Management Module is responsible for organizing, maintaining, and managing the educational curriculum structure within the ThinkPlus Adaptive Learning and Assessment Platform.

This module defines the hierarchical relationships between topics and subtopics, manages prerequisite dependencies, and provides the foundation for adaptive learning navigation, backtracking, and personalized learning path generation.

The Knowledge Graph serves as the curriculum intelligence layer of the platform. It represents how concepts are connected and determines the sequence in which students should learn topics.

The module shall support:

- Topic creation and management.
- Subtopic creation and management.
- Topic hierarchy management.
- Prerequisite relationship management.
- Knowledge graph construction.
- Learning path generation.
- Topic unlocking rules.
- Backtracking dependency management.
- Curriculum version management.
- Knowledge graph visualization.

The module shall act as the primary curriculum structure repository for:

- Adaptive Learning Engine
- Bayesian Knowledge Tracing (BKT) Engine
- Navigation and Backtracking Engine
- Assessment Management Module
- Analytics and Reporting Module

Priority

Priority Level: High

Rationale

The Knowledge Graph forms the structural backbone of the adaptive learning platform. Incorrect topic relationships can negatively impact learning progression, recommendation quality, and mastery tracking accuracy.

3.5.2 Stimulus / Response Sequences

Use Case 1: Create Topic

Stimulus

A content administrator creates a new topic.

Response

The system shall:

1. Validate topic information.
2. Assign a unique topic identifier.
3. Store topic metadata.
4. Add the topic to the curriculum structure.
5. Update the knowledge graph.

Use Case 2: Create Subtopic

Stimulus

A content administrator creates a new subtopic.

Response

The system shall:

1. Validate subtopic information.
2. Validate parent topic assignment.
3. Store subtopic metadata.
4. Update topic relationships.
5. Update the knowledge graph.

Use Case 3: Define Prerequisite Relationship

Stimulus

An administrator links Topic A as a prerequisite for Topic B.

Response

The system shall:

1. Validate topic relationships.
2. Check for cyclic dependencies.
3. Store prerequisite information.
4. Update the knowledge graph.
5. Validate curriculum integrity.

Use Case 4: Generate Learning Path

Stimulus

The Adaptive Learning Engine requests a learning path.

Response

The system shall:

1. Retrieve topic relationships.
2. Evaluate prerequisite conditions.
3. Determine eligible topics.
4. Generate an ordered learning path.
5. Return the recommended sequence.

Use Case 5: Backtracking Request

Stimulus

The Navigation Engine requests prerequisite topics for remediation.

Response

The system shall:

1. Identify prerequisite nodes.
2. Retrieve dependency paths.
3. Return recommended remediation topics.
4. Update navigation recommendations.

3.5.3 Functional Requirements

Topic Management

FR-KGM-001

The system shall allow authorized users to create topics.

FR-KGM-002

The system shall assign unique identifiers to topics.

FR-KGM-003

The system shall store topic metadata.

FR-KGM-004

The system shall allow modification of topic information.

FR-KGM-005

The system shall support topic archival.

Subtopic Management

FR-KGM-006

The system shall allow authorized users to create subtopics.

FR-KGM-007

The system shall associate each subtopic with a parent topic.

FR-KGM-008

The system shall maintain subtopic metadata.

FR-KGM-009

The system shall support modification of subtopic information.

FR-KGM-010

The system shall support subtopic archival.

Curriculum Hierarchy Management

FR-KGM-011

The system shall maintain hierarchical relationships between topics and subtopics.

FR-KGM-012

The system shall support multiple subtopics under a single topic.

FR-KGM-013

The system shall support curriculum organization by subject domain.

FR-KGM-014

The system shall maintain hierarchy integrity.

FR-KGM-015

The system shall prevent invalid hierarchy structures.

Knowledge Graph Management

FR-KGM-016

The system shall maintain a knowledge graph representing curriculum relationships.

FR-KGM-017

The system shall represent topics as graph nodes.

FR-KGM-018

The system shall represent prerequisite relationships as graph edges.

FR-KGM-019

The system shall maintain graph consistency.

FR-KGM-020

The system shall support graph updates when curriculum changes occur.

Prerequisite Management

FR-KGM-021

The system shall allow prerequisite relationships to be defined.

FR-KGM-022

The system shall support multiple prerequisites for a topic.

FR-KGM-023

The system shall validate prerequisite relationships.

FR-KGM-024

The system shall prevent circular dependencies.

FR-KGM-025

The system shall maintain prerequisite audit records.

Learning Path Management

FR-KGM-026

The system shall generate ordered learning paths.

FR-KGM-027

The system shall consider prerequisite dependencies during path generation.

FR-KGM-028

The system shall support adaptive learning path generation.

FR-KGM-029

The system shall support personalized curriculum progression.

FR-KGM-030

The system shall provide learning path recommendations.

Topic Unlocking Rules

FR-KGM-031

The system shall support topic unlocking mechanisms.

FR-KGM-032

The system shall evaluate prerequisite completion before unlocking topics.

FR-KGM-033

The system shall support mastery-based topic unlocking.

FR-KGM-034

The system shall support administrative override of unlocking conditions.

FR-KGM-035

The system shall maintain topic access records.

Backtracking Support

FR-KGM-036

The system shall support prerequisite retrieval for remediation purposes.

FR-KGM-037

The system shall identify prerequisite knowledge gaps.

FR-KGM-038

The system shall provide prerequisite topic recommendations.

FR-KGM-039

The system shall support adaptive remediation workflows.

FR-KGM-040

The system shall support future enhancement of dependency analysis.

Search and Navigation

FR-KGM-041

The system shall support topic search functionality.

FR-KGM-042

The system shall support subtopic search functionality.

FR-KGM-043

The system shall support navigation through topic relationships.

FR-KGM-044

The system shall support curriculum exploration.

FR-KGM-045

The system shall support knowledge graph navigation.

Curriculum Version Management

FR-KGM-046

The system shall maintain curriculum version information.

FR-KGM-047

The system shall maintain historical curriculum structures.

FR-KGM-048

The system shall support curriculum updates.

FR-KGM-049

The system shall preserve previous versions for auditing purposes.

FR-KGM-050

The system shall maintain change history records.

Integration Requirements

FR-KGM-051

The system shall provide topic information to the Assessment Management Module.

FR-KGM-052

The system shall provide topic and subtopic information to the Question Bank Management Module.

FR-KGM-053

The system shall provide prerequisite relationships to the Navigation & Backtracking Engine.

FR-KGM-054

The system shall provide topic structures to the Adaptive Learning Engine.

FR-KGM-055

The system shall provide curriculum metadata to Analytics and Reporting services.

Audit and Governance

FR-KGM-056

The system shall record topic creation activities.

FR-KGM-057

The system shall record topic modification activities.

FR-KGM-058

The system shall record prerequisite relationship changes.

FR-KGM-059

The system shall maintain curriculum audit trails.

FR-KGM-060

The system shall maintain governance records for compliance and reporting purposes.

3.6 Adaptive Learning Engine Module

3.6.1 Description and Priority

Description

The Adaptive Learning Engine is the core intelligence component of the ThinkPlus Adaptive Learning and Assessment Platform. This module is responsible for dynamically personalizing the learning experience for each student based on their learning behavior, mastery status, assessment performance, and estimated ability levels.

The Adaptive Learning Engine acts as the orchestration layer between multiple adaptive components including:

- Bayesian Knowledge Tracing (BKT) Engine
- IRT & EAP Ability Estimation Engine
- Fisher Information Question Selection Engine
- Navigation & Backtracking Engine
- Knowledge Graph Management Module
- Assessment Management Module

The engine continuously evaluates student interactions and determines the most appropriate learning action to maximize learning effectiveness and knowledge retention.

The Adaptive Learning Engine shall:

- Analyze student interactions.
- Evaluate learning progress.
- Coordinate adaptive decision making.
- Select personalized learning activities.
- Generate adaptive recommendations.
- Trigger remediation workflows.
- Trigger progression workflows.
- Coordinate question selection.
- Support personalized learning paths.

Priority

Priority Level: Critical

Rationale

The Adaptive Learning Engine represents the primary business differentiator of the ThinkPlus platform. Without this module, the platform functions as a traditional assessment system rather than an adaptive learning system.

3.6.2 Stimulus / Response Sequences

Use Case 1: Student Submits Question Response

Stimulus

A student submits a response during an active learning session.

Response

The system shall:

1. Validate the response.
2. Record interaction details.
3. Forward the interaction to the Adaptive Learning Engine.
4. Trigger mastery evaluation.
5. Trigger ability evaluation.
6. Determine the next learning action.
7. Request the next question.
8. Present the next learning activity.

Use Case 2: Student Completes Subtopic

Stimulus

The student's mastery level exceeds the defined mastery threshold.

Response

The system shall:

1. Mark the subtopic as mastered.
2. Evaluate eligible next subtopics.
3. Consult prerequisite rules.
4. Determine progression options.
5. Recommend the next learning activity.

Use Case 3: Student Struggles with a Subtopic

Stimulus

The student's mastery level falls below the minimum progression threshold.

Response

The system shall:

1. Detect the learning gap.
2. Evaluate prerequisite relationships.
3. Trigger remediation workflows.
4. Recommend prerequisite concepts.
5. Generate a revised learning path.

Use Case 4: New Learning Session Begins

Stimulus

A student starts a new learning session.

Response

The system shall:

1. Retrieve historical learning data.
2. Retrieve mastery states.
3. Retrieve ability estimates.
4. Determine the optimal starting point.
5. Generate a personalized learning session.

Use Case 5: Learning Path Recalculation

Stimulus

Significant changes occur in student performance or mastery states.

Response

The system shall:

1. Reevaluate learning progress.
2. Analyze curriculum dependencies.
3. Generate an updated learning path.
4. Update future recommendations.

3.6.3 Functional Requirements

Adaptive Decision Management

FR-ALE-001

The system shall evaluate student interactions in real time.

FR-ALE-002

The system shall generate adaptive learning decisions based on student performance.

FR-ALE-003

The system shall support individualized learning experiences.

FR-ALE-004

The system shall continuously update learning recommendations.

FR-ALE-005

The system shall determine appropriate learning actions after each interaction.

Learning State Evaluation

FR-ALE-006

The system shall evaluate student learning progress.

FR-ALE-007

The system shall evaluate mastery progression.

FR-ALE-008

The system shall evaluate estimated ability levels.

FR-ALE-009

The system shall identify learning strengths.

FR-ALE-010

The system shall identify learning weaknesses.

Personalization

FR-ALE-011

The system shall generate personalized learning paths.

FR-ALE-012

The system shall personalize learning recommendations.

FR-ALE-013

The system shall personalize question selection.

FR-ALE-014

The system shall personalize remediation activities.

FR-ALE-015

The system shall personalize progression activities.

Learning Path Management

FR-ALE-016

The system shall maintain dynamic learning paths.

FR-ALE-017

The system shall modify learning paths based on student performance.

FR-ALE-018

The system shall support prerequisite-aware progression.

FR-ALE-019

The system shall support mastery-based progression.

FR-ALE-020

The system shall support adaptive curriculum navigation.

Progression Management

FR-ALE-021

The system shall determine when students are ready to progress.

FR-ALE-022

The system shall recommend new learning content when progression conditions are met.

FR-ALE-023

The system shall support automatic topic unlocking.

FR-ALE-024

The system shall support adaptive progression rules.

FR-ALE-025

The system shall record progression decisions.

Remediation Management

FR-ALE-026

The system shall identify learning gaps.

FR-ALE-027

The system shall initiate remediation workflows.

FR-ALE-028

The system shall recommend prerequisite topics when necessary.

FR-ALE-029

The system shall support adaptive backtracking.

FR-ALE-030

The system shall maintain remediation history.

Recommendation Management

FR-ALE-031

The system shall generate learning recommendations.

FR-ALE-032

The system shall generate practice recommendations.

FR-ALE-033

The system shall generate revision recommendations.

FR-ALE-034

The system shall prioritize recommendations based on learning needs.

FR-ALE-035

The system shall support future recommendation enhancements.

Session Adaptation**FR-ALE-036**

The system shall adapt learning sessions dynamically.

FR-ALE-037

The system shall adjust content delivery during active sessions.

FR-ALE-038

The system shall support personalized session objectives.

FR-ALE-039

The system shall optimize question delivery sequences.

FR-ALE-040

The system shall maintain adaptive session histories.

Integration with BKT Engine**FR-ALE-041**

The system shall retrieve mastery information from the BKT Engine.

FR-ALE-042

The system shall use mastery estimates during adaptive decision making.

FR-ALE-043

The system shall evaluate subtopic mastery thresholds.

FR-ALE-044

The system shall support mastery-driven progression decisions.

FR-ALE-045

The system shall support mastery-driven remediation decisions.

Integration with IRT & EAP Engine**FR-ALE-046**

The system shall retrieve ability estimates from the IRT & EAP Engine.

FR-ALE-047

The system shall use ability estimates during question selection.

FR-ALE-048

The system shall support ability-based personalization.

FR-ALE-049

The system shall support topic-level ability evaluation.

FR-ALE-050

The system shall support future adaptive calibration enhancements.

Integration with Fisher Information Engine**FR-ALE-051**

The system shall request candidate questions from the Fisher Information Engine.

FR-ALE-052

The system shall use selected questions during adaptive sessions.

FR-ALE-053

The system shall support information-maximizing question selection.

FR-ALE-054

The system shall support difficulty-aware question delivery.

FR-ALE-055

The system shall support future question selection enhancements.

Analytics and Logging

FR-ALE-056

The system shall maintain adaptive decision logs.

FR-ALE-057

The system shall record recommendation history.

FR-ALE-058

The system shall record progression decisions.

FR-ALE-059

The system shall record remediation decisions.

FR-ALE-060

The system shall provide adaptive learning data to analytics and reporting services.

3.7 Bayesian Knowledge Tracing (BKT) Module

3.7.1 Description and Priority

Description

The Bayesian Knowledge Tracing (BKT) Module is responsible for estimating and tracking student mastery at the subtopic level throughout the learning process.

The module uses a probabilistic student model to estimate whether a student has mastered a specific subtopic based on observed interactions and assessment responses.

The BKT Engine shall maintain mastery estimates for each student-subtopic combination and continuously update those estimates after every learning interaction.

The module shall support:

- Subtopic mastery estimation.
- Mastery state tracking.
- Learning progression monitoring.
- Mastery threshold evaluation.

- Adaptive progression decisions.
- Adaptive remediation decisions.
- Learning history analysis.

The BKT Engine acts as the primary source of mastery information for the Adaptive Learning Engine.

Priority

Priority Level: Critical

Rationale

The BKT Engine provides the mastery signals used to determine whether students should progress, continue practicing, or receive remediation.

3.7.2 Stimulus / Response Sequences

Use Case 1: Student Answers Question

Stimulus

A student submits a response associated with a subtopic.

Response

The system shall:

1. Retrieve the student's current mastery state.
2. Evaluate response correctness.
3. Apply BKT update calculations.
4. Update mastery probability.
5. Store updated mastery values.
6. Notify the Adaptive Learning Engine.

Use Case 2: Mastery Threshold Reached

Stimulus

The student's mastery estimate exceeds the configured mastery threshold.

Response

The system shall:

1. Mark the subtopic as mastered.
2. Notify the Adaptive Learning Engine.
3. Trigger progression evaluation.
4. Update student learning records.

Use Case 3: Learning Difficulty Detected

Stimulus

Mastery estimates remain below progression thresholds.

Response

The system shall:

1. Identify learning gaps.
2. Trigger remediation workflows.
3. Recommend additional practice.
4. Notify the Adaptive Learning Engine.

3.7.3 Functional Requirements

Mastery Estimation

FR-BKT-001

The system shall maintain mastery estimates for each student-subtopic pair.

FR-BKT-002

The system shall update mastery estimates after every eligible interaction.

FR-BKT-003

The system shall support probabilistic mastery tracking.

FR-BKT-004

The system shall support configurable mastery thresholds.

FR-BKT-005

The system shall support future parameter recalibration.

State Management

FR-BKT-006

The system shall maintain historical mastery states.

FR-BKT-007

The system shall maintain mastery progression history.

FR-BKT-008

The system shall support retrieval of current mastery values.

FR-BKT-009

The system shall support storage of mastery transitions.

FR-BKT-010

The system shall maintain audit logs for mastery updates.

Progression Support**FR-BKT-011**

The system shall support mastery-based progression.

FR-BKT-012

The system shall support mastery-based remediation.

FR-BKT-013

The system shall support adaptive learning decisions.

FR-BKT-014

The system shall notify the Adaptive Learning Engine when mastery thresholds are crossed.

FR-BKT-015

The system shall support future multi-skill mastery extensions.

Analytics Support**FR-BKT-016**

The system shall provide mastery analytics.

FR-BKT-017

The system shall provide subtopic mastery reports.

FR-BKT-018

The system shall support historical trend analysis.

FR-BKT-019

The system shall expose mastery data to reporting modules.

The system shall maintain mastery-related audit records.

3.8 IRT & EAP Ability Estimation Module

3.8.1 Description and Priority

Description

The IRT & EAP Ability Estimation Module is responsible for estimating a student's ability level at the topic level using Item Response Theory (IRT) and Expected A Posteriori (EAP) estimation techniques.

The module provides a continuous ability score (0) representing the student's current capability within a specific topic.

The estimated ability value shall be used by the Adaptive Learning Engine and Fisher Information Question Selection Engine to select appropriate questions and personalize learning experiences.

The module shall support:

- Topic-level ability estimation.
- Continuous ability updates.
- EAP-based ability calculations.
- Ability history tracking.
- Confidence estimation.
- Ability analytics and reporting.

The module does not determine mastery. Mastery estimation remains the responsibility of the BKT Engine.

Priority

Priority Level: Critical

Rationale

The IRT & EAP Engine provides the psychometric foundation for adaptive question difficulty selection and student ability measurement.

3.8.2 Stimulus / Response Sequences

Use Case 1: Student Answers Question

Stimulus

A student submits a response to a question.

Response

The system shall:

1. Retrieve question parameters.
2. Retrieve current topic ability estimate.
3. Execute EAP estimation.
4. Calculate updated ability score.
5. Store updated ability information.
6. Notify the Adaptive Learning Engine.

Use Case 2: Ability Recalculation

Stimulus

Additional assessment evidence becomes available.

Response

The system shall:

1. Retrieve historical response data.
2. Recalculate ability estimates.
3. Update topic-level ability values.
4. Store revised estimates.

Use Case 3: Question Selection Request

Stimulus

The Fisher Information Engine requests ability estimates.

Response

The system shall:

1. Retrieve current ability values.
2. Return topic-level ability estimates.
3. Return associated confidence metrics.

3.8.3 Functional Requirements

Ability Estimation

FR-IRT-001

The system shall maintain topic-level ability estimates.

FR-IRT-002

The system shall calculate ability values using Item Response Theory.

FR-IRT-003

The system shall support Expected A Posteriori estimation.

FR-IRT-004

The system shall update ability estimates after eligible interactions.

FR-IRT-005

The system shall support continuous ability tracking.

Ability State Management

FR-IRT-006

The system shall store historical ability values.

FR-IRT-007

The system shall maintain ability progression history.

FR-IRT-008

The system shall support retrieval of current ability estimates.

FR-IRT-009

The system shall maintain confidence information.

FR-IRT-010

The system shall support future recalibration processes.

Topic-Level Measurement

FR-IRT-011

The system shall maintain separate ability estimates for each topic.

FR-IRT-012

The system shall support ability estimation across multiple topics.

FR-IRT-013

The system shall support topic-specific performance tracking.

FR-IRT-014

The system shall support topic comparison analytics.

FR-IRT-015

The system shall maintain topic-level audit history.

Adaptive Learning Support**FR-IRT-016**

The system shall provide ability estimates to the Adaptive Learning Engine.

FR-IRT-017

The system shall support personalized question selection.

FR-IRT-018

The system shall support difficulty matching.

FR-IRT-019

The system shall support adaptive learning recommendations.

FR-IRT-020

The system shall support future adaptive enhancements.

Fisher Information Integration**FR-IRT-021**

The system shall provide ability estimates to the Fisher Information Engine.

FR-IRT-022

The system shall support information-maximizing question selection.

FR-IRT-023

The system shall support difficulty-aware recommendations.

FR-IRT-024

The system shall support adaptive content delivery.

FR-IRT-025

The system shall support future psychometric model upgrades.

Reporting and Analytics

FR-IRT-026

The system shall provide topic-level ability reports.

FR-IRT-027

The system shall provide ability progression reports.

FR-IRT-028

The system shall support historical ability trend analysis.

FR-IRT-029

The system shall expose ability data to analytics services.

FR-IRT-030

The system shall maintain audit logs for ability estimation activities.

3.9 Fisher Information Question Selection Module

3.9.1 Description and Priority

Description

The Fisher Information Question Selection Module is responsible for selecting the most appropriate question for a student during adaptive learning and adaptive assessments.

The module utilizes student ability estimates provided by the IRT & EAP Ability Estimation Engine and candidate question pools provided by the Adaptive Learning Engine.

The objective of this module is to maximize information gain from each student interaction while maintaining an appropriate difficulty level.

The module shall:

- Evaluate candidate questions.
- Calculate information values.
- Rank candidate questions.
- Select the most informative question.
- Support adaptive difficulty matching.
- Improve ability estimation efficiency.
- Optimize assessment precision.

The Fisher Information Engine does not determine learning paths or mastery progression. It only determines which question should be presented from an eligible question pool.

Priority

Priority Level: Critical

Rationale

Efficient question selection is essential for reducing assessment length while improving measurement accuracy and adaptive learning quality.

3.9.2 Stimulus / Response Sequences

Use Case 1: Question Selection Request

Stimulus

The Adaptive Learning Engine requests the next question.

Response

The system shall:

1. Retrieve student ability estimate.
2. Retrieve candidate question pool.
3. Calculate information values.
4. Rank candidate questions.
5. Select the optimal question.
6. Return the selected question.

Use Case 2: Ability Update

Stimulus

A student's ability estimate changes.

Response

The system shall:

1. Retrieve updated ability values.
2. Recalculate information values.
3. Update question rankings.
4. Prepare future recommendations.

Use Case 3: Limited Question Pool

Stimulus

The candidate pool contains insufficient questions.

Response

The system shall:

1. Apply fallback selection rules.
2. Expand candidate search scope.
3. Select the best available question.
4. Notify the Adaptive Learning Engine.

3.9.3 Functional Requirements

Question Evaluation

FR-FIS-001

The system shall evaluate candidate questions before selection.

FR-FIS-002

The system shall calculate information values for candidate questions.

FR-FIS-003

The system shall rank candidate questions.

FR-FIS-004

The system shall select the highest-ranked eligible question.

FR-FIS-005

The system shall support future ranking enhancements.

Ability-Based Selection

FR-FIS-006

The system shall use topic-level ability estimates during question selection.

FR-FIS-007

The system shall support adaptive difficulty matching.

FR-FIS-008

The system shall support personalized question delivery.

FR-FIS-009

The system shall support efficient ability measurement.

FR-FIS-010

The system shall support future psychometric upgrades.

Candidate Pool Management

FR-FIS-011

The system shall retrieve eligible question pools.

FR-FIS-012

The system shall exclude previously used questions when required.

FR-FIS-013

The system shall support difficulty filtering.

FR-FIS-014

The system shall support topic filtering.

FR-FIS-015

The system shall support adaptive pool expansion.

Integration Requirements

FR-FIS-016

The system shall receive ability estimates from the IRT & EAP Engine.

FR-FIS-017

The system shall receive candidate questions from the Adaptive Learning Engine.

FR-FIS-018

The system shall provide selected questions to the Adaptive Learning Engine.

FR-FIS-019

The system shall support adaptive assessment workflows.

FR-FIS-020

The system shall support learning session workflows.

Analytics and Audit**FR-FIS-021**

The system shall maintain question selection history.

FR-FIS-022

The system shall record information values.

FR-FIS-023

The system shall record selection decisions.

FR-FIS-024

The system shall support reporting requirements.

FR-FIS-025

The system shall maintain audit logs.

3.10 Navigation & Backtracking Module

3.10.1 Description and Priority

Description

The Navigation & Backtracking Module is responsible for determining student movement through the curriculum based on mastery levels, prerequisite relationships, learning gaps, and adaptive learning decisions.

This module uses information from:

- Knowledge Graph Management Module
- Bayesian Knowledge Tracing (BKT) Module
- Adaptive Learning Engine
- Assessment Management Module

The Navigation & Backtracking Module ensures students progress through learning content in an appropriate sequence while preventing advancement when prerequisite knowledge gaps exist.

The module shall:

- Manage topic progression.
- Manage subtopic progression.
- Support prerequisite validation.
- Support adaptive navigation.
- Support remediation workflows.
- Support curriculum backtracking.
- Generate personalized learning routes.

Priority

Priority Level: Critical

Rationale

This module ensures that students learn concepts in the correct order and receive remediation when prerequisite knowledge gaps are detected.

3.10.2 Stimulus / Response Sequences

Use Case 1: Subtopic Mastered

Stimulus

The BKT Engine reports mastery achievement.

Response

The system shall:

1. Evaluate progression conditions.

2. Check prerequisite requirements.
3. Identify eligible next subtopics.
4. Generate progression recommendations.
5. Notify the Adaptive Learning Engine.

Use Case 2: Learning Gap Detected

Stimulus

Mastery remains below progression thresholds.

Response

The system shall:

1. Identify prerequisite dependencies.
2. Locate knowledge gaps.
3. Generate remediation paths.
4. Recommend prerequisite learning activities.

Use Case 3: Topic Completion

Stimulus

A student completes all required subtopics within a topic.

Response

The system shall:

1. Mark the topic as completed.
2. Evaluate successor topics.
3. Apply unlocking rules.
4. Recommend future learning activities.

Use Case 4: Adaptive Navigation Request

Stimulus

The Adaptive Learning Engine requests navigation guidance.

Response

The system shall:

1. Analyze mastery information.
2. Analyze prerequisite relationships.
3. Generate navigation recommendations.
4. Return recommended learning paths.

3.10.3 Functional Requirements

Progression Management

FR-NAV-001

The system shall support progression between subtopics.

FR-NAV-002

The system shall support progression between topics.

FR-NAV-003

The system shall support mastery-based progression.

FR-NAV-004

The system shall support adaptive progression decisions.

FR-NAV-005

The system shall maintain progression history.

Backtracking Management

FR-NAV-006

The system shall support prerequisite-based backtracking.

FR-NAV-007

The system shall identify prerequisite learning gaps.

FR-NAV-008

The system shall recommend prerequisite learning activities.

FR-NAV-009

The system shall support adaptive remediation.

FR-NAV-010

The system shall maintain remediation history.

Knowledge Graph Integration

FR-NAV-011

The system shall retrieve prerequisite relationships from the Knowledge Graph.

FR-NAV-012

The system shall support graph-based navigation.

FR-NAV-013

The system shall support dependency analysis.

FR-NAV-014

The system shall support curriculum path generation.

FR-NAV-015

The system shall support future graph enhancements.

Topic Unlocking

FR-NAV-016

The system shall support topic unlocking rules.

FR-NAV-017

The system shall evaluate progression eligibility.

FR-NAV-018

The system shall support mastery-based unlocking.

FR-NAV-019

The system shall support administrator overrides.

FR-NAV-020

The system shall maintain unlocking history.

Learning Path Generation

FR-NAV-021

The system shall generate personalized learning paths.

FR-NAV-022

The system shall generate remediation paths.

FR-NAV-023

The system shall generate progression paths.

FR-NAV-024

The system shall support adaptive curriculum navigation.

FR-NAV-025

The system shall support future recommendation enhancements.

Analytics and Audit

FR-NAV-026

The system shall record navigation decisions.

FR-NAV-027

The system shall record progression events.

FR-NAV-028

The system shall record remediation events.

FR-NAV-029

The system shall support reporting requirements.

FR-NAV-030

The system shall maintain navigation audit logs.

3.11 Analytics & Reporting Module

3.11.1 Description and Priority

Description

The Analytics & Reporting Module is responsible for collecting, processing, analyzing, and presenting educational, assessment, mastery, and ability-related data generated throughout the platform.

The module provides actionable insights to students, teachers, administrators, and management users through dashboards, reports, visualizations, and performance metrics.

The module shall aggregate information from:

- Student Learning Module

- Assessment Management Module
- BKT Engine
- IRT & EAP Engine
- Adaptive Learning Engine
- Navigation & Backtracking Module

The Analytics & Reporting Module shall support:

- Student performance reporting.
- Topic-level analytics.
- Subtopic-level analytics.
- Mastery reporting.
- Ability progression reporting.
- Assessment analytics.
- Learning trend analysis.
- Administrative reporting.
- Data export functionality.

Priority

Priority Level: High

Rationale

Analytics and reporting provide visibility into learning effectiveness and support educational decision-making.

3.11.2 Stimulus / Response Sequences

Use Case 1: Student Views Performance Report

Stimulus

A student requests performance analytics.

Response

The system shall:

1. Retrieve student performance data.
2. Calculate analytics metrics.
3. Generate performance reports.
4. Display visualizations and summaries.

Use Case 2: Teacher Reviews Student Progress

Stimulus

A teacher requests progress information.

Response

The system shall:

1. Retrieve student records.
2. Retrieve mastery information.
3. Retrieve ability estimates.
4. Generate progress reports.

Use Case 3: Administrator Generates Platform Report**Stimulus**

An administrator requests organizational analytics.

Response

The system shall:

1. Aggregate platform metrics.
2. Generate summary reports.
3. Provide export options.

3.11.3 Functional Requirements

FR-ANL-001

The system shall provide student performance reports.

FR-ANL-002

The system shall provide topic-level analytics.

FR-ANL-003

The system shall provide subtopic-level analytics.

FR-ANL-004

The system shall provide mastery reports.

FR-ANL-005

The system shall provide ability progression reports.

FR-ANL-006

The system shall support assessment analytics.

FR-ANL-007

The system shall support learning trend analysis.

FR-ANL-008

The system shall support dashboard visualizations.

FR-ANL-009

The system shall support report exports.

FR-ANL-010

The system shall maintain reporting audit logs.

3.12 Teacher Portal Module

3.12.1 Description and Priority

Description

The Teacher Portal Module provides educators with tools to monitor student progress, analyze learning outcomes, review mastery levels, track ability estimates, and support student learning activities.

The Teacher Portal acts as the primary interface between educators and the adaptive learning platform.

The module shall support:

- Student monitoring.
- Performance tracking.
- Mastery analysis.
- Ability analysis.
- Learning intervention support.
- Assessment review.
- Progress reporting.
- Class-level analytics.

Priority

Priority Level: High

Rationale

Teachers require visibility into student performance to provide effective educational support and intervention.

3.12.2 Stimulus / Response Sequences

Use Case 1: Teacher Views Student Performance

Stimulus

A teacher selects a student.

Response

The system shall:

1. Retrieve learning history.
2. Retrieve mastery information.
3. Retrieve ability estimates.
4. Generate performance dashboards.

Use Case 2: Teacher Reviews Class Performance

Stimulus

A teacher requests class analytics.

Response

The system shall:

1. Aggregate student metrics.
2. Generate class reports.
3. Display performance summaries.

Use Case 3: Teacher Identifies Learning Gaps

Stimulus

A teacher requests learning gap analysis.

Response

The system shall:

1. Analyze mastery information.
2. Identify weak areas.
3. Generate intervention recommendations.

3.12.3 Functional Requirements

FR-TCH-001

The system shall provide teacher dashboards.

FR-TCH-002

The system shall allow teachers to view student progress.

FR-TCH-003

The system shall provide mastery reports.

FR-TCH-004

The system shall provide ability reports.

FR-TCH-005

The system shall provide assessment results.

FR-TCH-006

The system shall provide class-level analytics.

FR-TCH-007

The system shall support learning gap identification.

FR-TCH-008

The system shall support intervention planning.

FR-TCH-009

The system shall support report exports.

FR-TCH-010

The system shall maintain teacher activity logs.

3.13 Administrator Portal Module

3.13.1 Description and Priority

Description

The Administrator Portal Module provides platform administrators with tools for system management, user administration, content governance, configuration management, analytics oversight, and operational monitoring.

The module serves as the central management interface for the ThinkPlus Adaptive Learning and Assessment Platform.

The module shall support:

- User management.
- Role management.
- Topic management.
- Knowledge graph management.
- Question bank governance.
- System configuration.
- Platform monitoring.
- Reporting and analytics oversight.
- Audit management.

Priority

Priority Level: Critical

Rationale

Administrative capabilities are essential for maintaining system integrity, governance, security, and operational stability.

3.13.2 Stimulus / Response Sequences

Use Case 1: User Administration

Stimulus

An administrator creates or modifies a user account.

Response

The system shall:

1. Validate administrative permissions.
2. Apply requested changes.
3. Record audit information.
4. Notify relevant services.

Use Case 2: Curriculum Administration

Stimulus

An administrator updates topics, subtopics, or knowledge graph structures.

Response

The system shall:

1. Validate changes.
2. Update curriculum information.
3. Maintain version history.
4. Notify dependent services.

Use Case 3: System Monitoring

Stimulus

An administrator requests system health information.

Response

The system shall:

1. Retrieve operational metrics.
2. Generate monitoring dashboards.
3. Display platform status information.

3.13.3 Functional Requirements

User Administration

FR-ADM-001

The system shall allow administrators to create user accounts.

FR-ADM-002

The system shall allow administrators to update user accounts.

FR-ADM-003

The system shall allow administrators to activate and deactivate accounts.

FR-ADM-004

The system shall allow administrators to assign user roles.

FR-ADM-005

The system shall maintain user administration audit logs.

Content Administration

FR-ADM-006

The system shall allow administrators to manage topics.

FR-ADM-007

The system shall allow administrators to manage subtopics.

FR-ADM-008

The system shall allow administrators to manage knowledge graph structures.

FR-ADM-009

The system shall allow administrators to manage question banks.

FR-ADM-010

The system shall support curriculum governance workflows.

System Configuration

FR-ADM-011

The system shall allow configuration of platform settings.

FR-ADM-012

The system shall allow configuration of adaptive learning parameters.

FR-ADM-013

The system shall allow configuration of assessment settings.

FR-ADM-014

The system shall allow configuration of reporting settings.

FR-ADM-015

The system shall maintain configuration history.

Monitoring and Governance

FR-ADM-016

The system shall provide operational dashboards.

FR-ADM-017

The system shall provide platform usage analytics.

FR-ADM-018

The system shall provide audit reports.

FR-ADM-019

The system shall support compliance reporting.

FR-ADM-020

The system shall maintain administrative audit logs.

4. External Interface Requirements

4.1 User Interfaces

Overview

The ThinkPlus Adaptive Learning and Assessment Platform shall provide responsive web-based user interfaces accessible through modern web browsers.

The user interface shall support role-based experiences for:

- Students
- Teachers
- Content Managers
- Administrators
- Management Users

The user interface shall be designed to provide an intuitive, accessible, and consistent user experience.

Student Interface

The Student Interface shall provide:

- Secure login and authentication.
- Personalized dashboard.
- Learning session interface.
- Assessment interface.
- Progress tracking dashboards.
- Performance reports.
- Learning recommendations.
- Topic and subtopic navigation.

UI-001

The system shall provide a responsive student dashboard.

UI-002

The system shall display learning progress indicators.

UI-003

The system shall display assessment results.

UI-004

The system shall display adaptive learning recommendations.

UI-005

The system shall support desktop, tablet, and mobile devices.

Teacher Interface

The Teacher Interface shall provide:

- Student monitoring dashboards.
- Class performance reports.
- Learning analytics.
- Assessment review capabilities.
- Progress tracking tools.

UI-006

The system shall provide teacher-specific dashboards.

UI-007

The system shall provide student performance visualizations.

UI-008

The system shall provide class-level analytics.

Administrator Interface

The Administrator Interface shall provide:

- User management.
- Role management.
- Topic management.
- Question bank management.
- System configuration.

- Analytics monitoring.

UI-009

The system shall provide administrative management interfaces.

UI-010

The system shall provide system monitoring dashboards.

Accessibility

UI-011

The system shall support modern accessibility standards.

UI-012

The system shall provide readable typography and layouts.

UI-013

The system shall support keyboard navigation where applicable.

4.2 Hardware Interfaces

Overview

The platform is designed as a cloud-hosted web application and shall not require specialized hardware.

Supported client devices include:

- Desktop computers
- Laptop computers
- Tablets
- Mobile devices

Client Hardware Requirements

HW-001

The system shall support standard desktop and laptop hardware.

HW-002

The system shall support modern mobile devices.

HW-003

The system shall support touchscreen-enabled devices.

HW-004

The system shall require internet connectivity for operation.

Server Infrastructure

HW-005

The system shall support deployment on cloud-hosted infrastructure.

HW-006

The system shall support scalable compute resources.

HW-007

The system shall support scalable storage resources.

4.3 Software Interfaces

Overview

The platform shall integrate with internal and external software components through defined interfaces and APIs.

Frontend Interface

Technology Stack:

- Next.js
- TypeScript

SW-001

The frontend shall communicate with backend services through REST APIs.

SW-002

The frontend shall support secure token-based authentication.

SW-003

The frontend shall support responsive rendering.

Backend Interface

Technology Stack:

- Python
- FastAPI

SW-004

The backend shall expose RESTful APIs.

SW-005

The backend shall validate requests and responses.

SW-006

The backend shall support secure API access.

Database Interface

Technology Stack:

- PostgreSQL

SW-007

The system shall persist operational data in PostgreSQL.

SW-008

The system shall support transactional consistency.

SW-009

The system shall support concurrent access.

Authentication Interface

Technology Stack:

- JWT Authentication
- bcrypt Password Hashing

SW-010

The system shall support token-based authentication.

SW-011

The system shall securely store password credentials.

SW-012

The system shall validate user authorization before granting access.

Analytics Interface

SW-013

The system shall provide analytics data interfaces.

SW-014

The system shall provide reporting service interfaces.

4.4 Communications Interfaces

Overview

The platform shall support secure communication between client devices, application services, and data storage systems.

Web Communications

COM-001

The system shall use HTTPS for all client-server communications.

COM-002

The system shall encrypt sensitive information during transmission.

COM-003

The system shall support secure session management.

API Communications

COM-004

The system shall support REST-based API communication.

COM-005

The system shall support JSON request and response payloads.

COM-006

The system shall validate incoming API requests.

COM-007

The system shall return standardized API responses.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Overview

The ThinkPlus Adaptive Learning and Assessment Platform shall provide responsive, reliable, and efficient performance under expected operational loads.

The platform shall be designed to support real-time adaptive learning workflows without noticeable delays to end users.

Performance Requirements

PERF-001

The system shall respond to user interface requests

PERF-002

The system shall authenticate users

PERF-003

The system shall generate adaptive learning decisions after a response is submitted.

PERF-004

The system shall retrieve assessment questions

PERF-005

The system shall process assessment submissions

PERF-006

The BKT Engine shall update mastery

PERF-007

The IRT & EAP Engine shall update and estimates ability

PERF-008

The Fisher Information Engine shall select candidate questions.

5.2 Security Requirements

Overview

The platform shall implement industry-standard security controls to protect user identities, learning records, assessment content, and administrative functions.

Authentication Security

SEC-001

The system shall require authentication before accessing protected resources.

SEC-002

The system shall implement Role-Based Access Control (RBAC).

SEC-003

The system shall restrict access based on assigned permissions.

SEC-004

The system shall support secure logout functionality.

Credential Security

SEC-005

The system shall store passwords using bcrypt hashing algorithms.

SEC-006

The system shall never store plaintext passwords.

SEC-007

The system shall enforce password complexity requirements.

SEC-008

The system shall support secure password reset workflows.

API Security

SEC-009

The system shall use JWT-based authentication tokens.

SEC-010

The system shall validate authentication tokens before processing requests.

SEC-011

The system shall reject unauthorized API requests.

SEC-012

The system shall implement API authorization controls.

Data Security

SEC-013

The system shall encrypt data in transit using HTTPS/TLS.

SEC-014

The system shall protect sensitive user information.

SEC-015

The system shall restrict direct database access.

SEC-016

The system shall support audit logging of security events.

SEC-017

The system shall maintain security incident records.

5.4 Software Quality Attributes

Usability

QUAL-010

The system shall provide intuitive user interfaces.

QUAL-011

The system shall provide clear navigation structures.

QUAL-012

The system shall provide understandable error messages.

Portability**QUAL-013**

The system shall support deployment across cloud platforms.

QUAL-014

The system shall support modern web browsers.

QUAL-015

The system shall support containerized deployment environments.

6. Other Requirements

6.1 Database Requirements

Overview

The ThinkPlus Adaptive Learning and Assessment Platform shall utilize PostgreSQL as the primary transactional database for storing operational, educational, assessment, user, and adaptive learning data.

The database architecture shall support scalability, consistency, integrity, and high-performance query execution.

Core Database Requirements**DB-001**

The system shall store user account information.

DB-002

The system shall store role and permission information.

DB-003

The system shall store topic and subtopic structures.

DB-004

The system shall store knowledge graph relationships.

DB-005

The system shall store question bank information.

DB-006

The system shall store assessment information.

DB-007

The system shall store student responses.

DB-008

The system shall store learning session information.

DB-009

The system shall store BKT mastery states.

DB-010

The system shall store IRT ability estimates.

Data Integrity Requirements

DB-011

The system shall enforce referential integrity.

DB-012

The system shall enforce primary key constraints.

DB-013

The system shall enforce foreign key constraints.

DB-014

The system shall support transactional consistency.

DB-015

The system shall prevent duplicate records where applicable.

Database Performance Requirements**DB-016**

The system shall support indexed queries for frequently accessed data.

DB-017

The system shall optimize retrieval of student learning records.

DB-018

The system shall optimize retrieval of question bank records.

DB-019

The system shall optimize retrieval of assessment results.

DB-020

The system shall support future database partitioning strategies.

Backup and Recovery Requirements**DB-021**

The system shall support scheduled backups.

DB-022

The system shall support point-in-time recovery.

DB-023

The system shall support disaster recovery procedures.

DB-024

The system shall maintain backup retention policies.

DB-025

The system shall support restoration testing procedures.

6.2 Logging Requirements

Overview

The platform shall maintain comprehensive audit and operational logs to support monitoring, debugging, compliance, analytics, and security investigations.

Logging shall be implemented across all major modules.

Application Logging

LOG-001

The system shall record application startup events.

LOG-002

The system shall record application shutdown events.

LOG-003

The system shall record application exceptions.

LOG-004

The system shall record API requests and responses.

LOG-005

The system shall record system warnings and errors.

User Activity Logging

LOG-006

The system shall record user login events.

LOG-007

The system shall record logout events.

LOG-008

The system shall record password changes.

LOG-009

The system shall record profile updates.

LOG-010

The system shall record user activity history.

Assessment Logging

LOG-011

The system shall record assessment start events.

LOG-012

The system shall record assessment completion events.

LOG-013

The system shall record student responses.

LOG-014

The system shall record assessment scores.

LOG-015

The system shall record assessment modifications.

Adaptive Learning Logging

LOG-016

The system shall record adaptive learning decisions.

LOG-017

The system shall record BKT state transitions.

LOG-018

The system shall record IRT ability updates.

LOG-019

The system shall record Fisher Information selection decisions.

LOG-020

The system shall record navigation and backtracking decisions.

Security Logging

LOG-021

The system shall record failed login attempts.

LOG-022

The system shall record unauthorized access attempts.

LOG-023

The system shall record permission changes.

LOG-024

The system shall record administrative actions.

LOG-025

The system shall maintain security audit trails.

Log Retention

LOG-026

The system shall maintain operational logs for a configurable retention period.

LOG-027

The system shall support archival of historical logs.

LOG-028

The system shall support log search functionality.

LOG-029

The system shall support log export functionality.

LOG-030

The system shall support future SIEM integrations.

6.3 AI and Adaptive Learning Requirements

Overview

The ThinkPlus platform shall implement an adaptive learning architecture combining psychometric measurement, mastery estimation, curriculum navigation, and intelligent question selection.

The adaptive learning framework shall utilize:

- Bayesian Knowledge Tracing (BKT)
- Item Response Theory (IRT)
- Expected A Posteriori (EAP)
- Fisher Information Question Selection
- Knowledge Graph Navigation
- Adaptive Backtracking

Bayesian Knowledge Tracing Requirements

AI-001

The system shall estimate mastery at the subtopic level.

AI-002

The system shall maintain mastery probabilities for each student-subtopic pair.

AI-003

The system shall update mastery estimates after eligible interactions.

AI-004

The system shall support configurable mastery thresholds.

AI-005

The system shall support mastery-based progression decisions.

IRT & EAP Requirements

AI-006

The system shall estimate topic-level ability values.

AI-007

The system shall support EAP-based ability estimation.

AI-008

The system shall maintain topic-specific ability estimates.

AI-009

The system shall support adaptive difficulty matching.

AI-010

The system shall support future psychometric recalibration processes.

Fisher Information Requirements

AI-011

The system shall select questions from eligible candidate pools.

AI-012

The system shall prioritize highly informative questions.

AI-013

The system shall support ability-based question selection.

AI-014

The system shall support adaptive assessment workflows.

AI-015

The system shall support adaptive learning workflows.

Knowledge Graph Requirements

AI-016

The system shall maintain prerequisite relationships between topics and subtopics.

AI-017

The system shall support graph-based learning progression.

AI-018

The system shall support graph-based remediation.

AI-019

The system shall support adaptive curriculum navigation.

AI-020

The system shall support future graph expansion.

Adaptive Navigation Requirements

AI-021

The system shall support mastery-based progression.

AI-022

The system shall support prerequisite-aware navigation.

AI-023

The system shall support adaptive backtracking.

AI-024

The system shall support remediation workflows.

AI-025

The system shall support personalized learning paths.

Future Enhancement Requirements

AI-026

The system architecture shall support future 2PL IRT upgrades.

AI-027

The system architecture shall support future MIRT integration.

AI-028

The system architecture shall support future Half-Life Regression (HLR) integration.

AI-029

The system architecture shall support future adaptive recommendation enhancements.

The system architecture shall support future machine learning model integration.

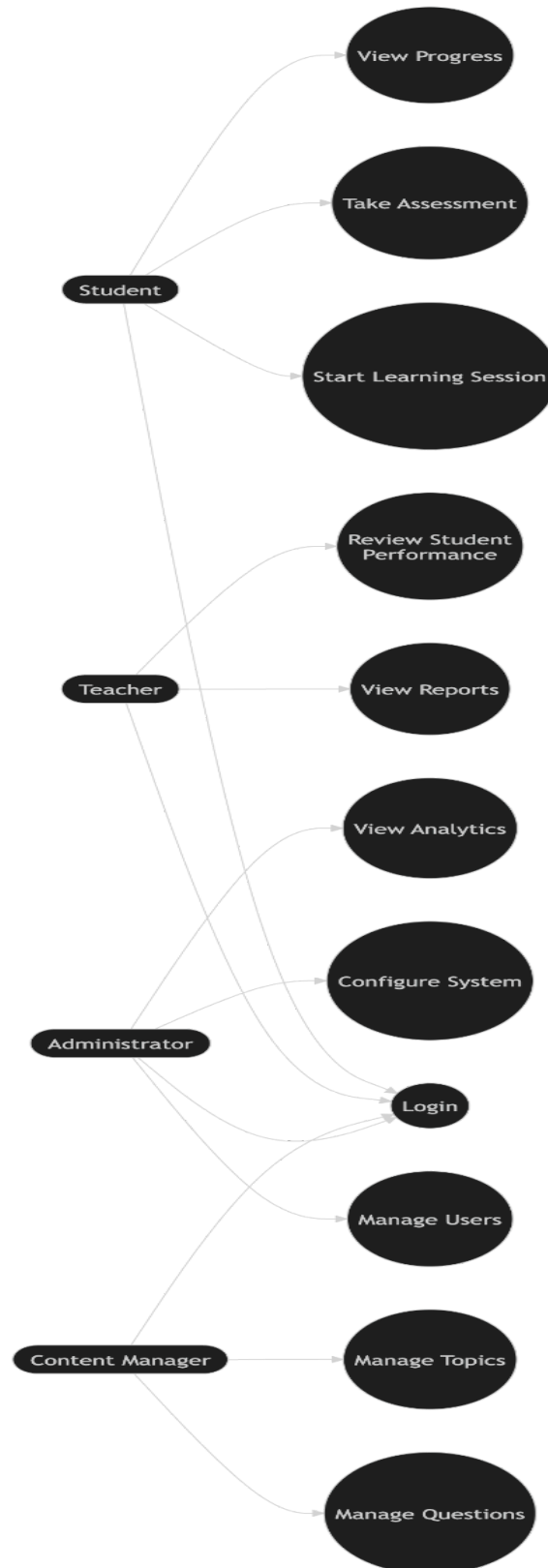
Appendix A: Glossary

Term	Definition
Adaptive Learning	A learning approach that dynamically adjusts content, difficulty, and learning paths based on student performance.
Assessment	A structured evaluation used to measure student knowledge and performance.
Ability (θ)	A latent trait estimate representing a student's capability within a topic.
Bayesian Knowledge Tracing (BKT)	A probabilistic model used to estimate mastery of individual subtopics.
EAP (Expected A Posteriori)	A Bayesian method used to estimate student ability in Item Response Theory.
Fisher Information	A statistical measure used to determine which question provides the most information about a student's ability.
IRT (Item Response Theory)	A psychometric framework used to estimate student ability and question difficulty relationships.
Knowledge Graph	A graph-based curriculum representation showing relationships between topics and prerequisites.

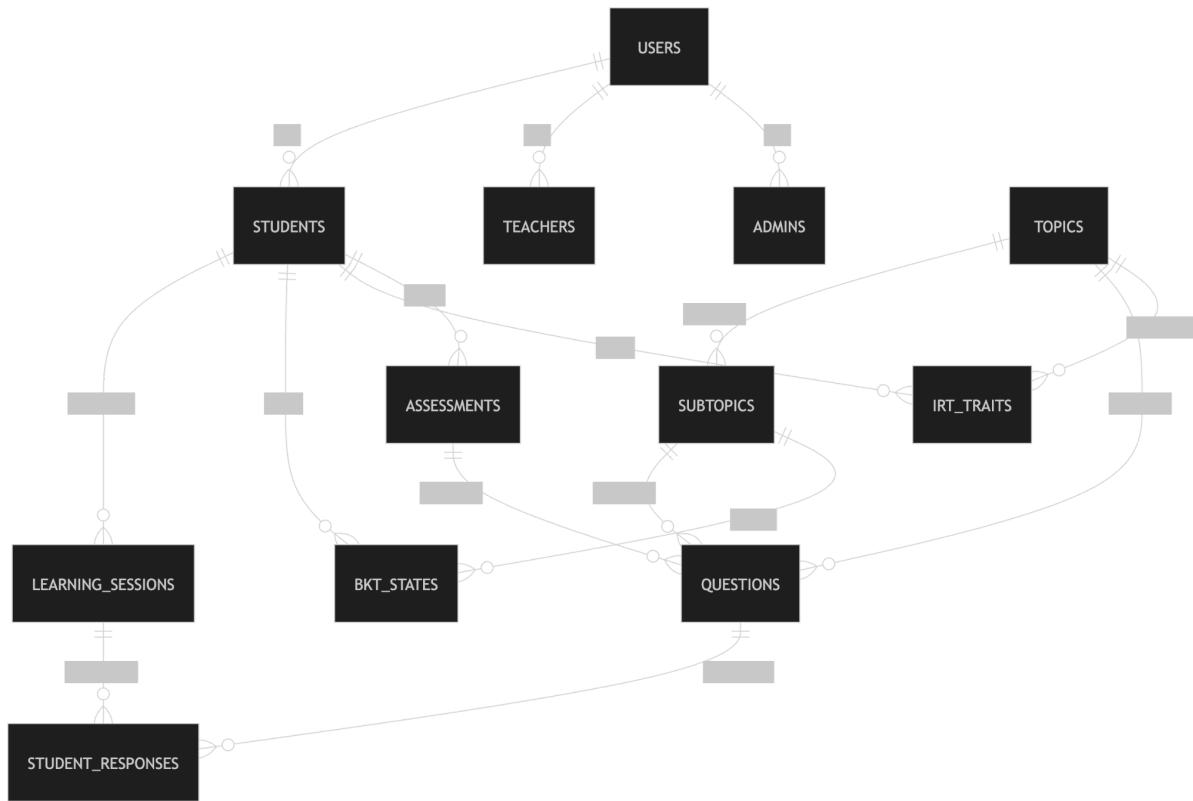
Mastery	The estimated probability that a student has learned a specific subtopic.
Question Bank	Repository containing all assessment and learning questions.
Subtopic	A granular learning concept belonging to a topic.
Topic	A major learning concept within a subject domain.
Remediation	Additional learning activities assigned to address learning gaps.
Adaptive Engine	Core system responsible for making personalized learning decisions.
JWT	JSON Web Token used for secure authentication and authorization.
RBAC	Role-Based Access Control mechanism used for authorization.
PostgreSQL	Primary transactional database used by the platform.
FastAPI	Backend application framework used for API development.
Next.js	Frontend framework used for web application development.
ClickHouse	Future analytical database intended for large-scale reporting workloads.
HLR	Half-Life Regression, a future spaced repetition model for memory retention optimization.

Appendix B: Analysis Models

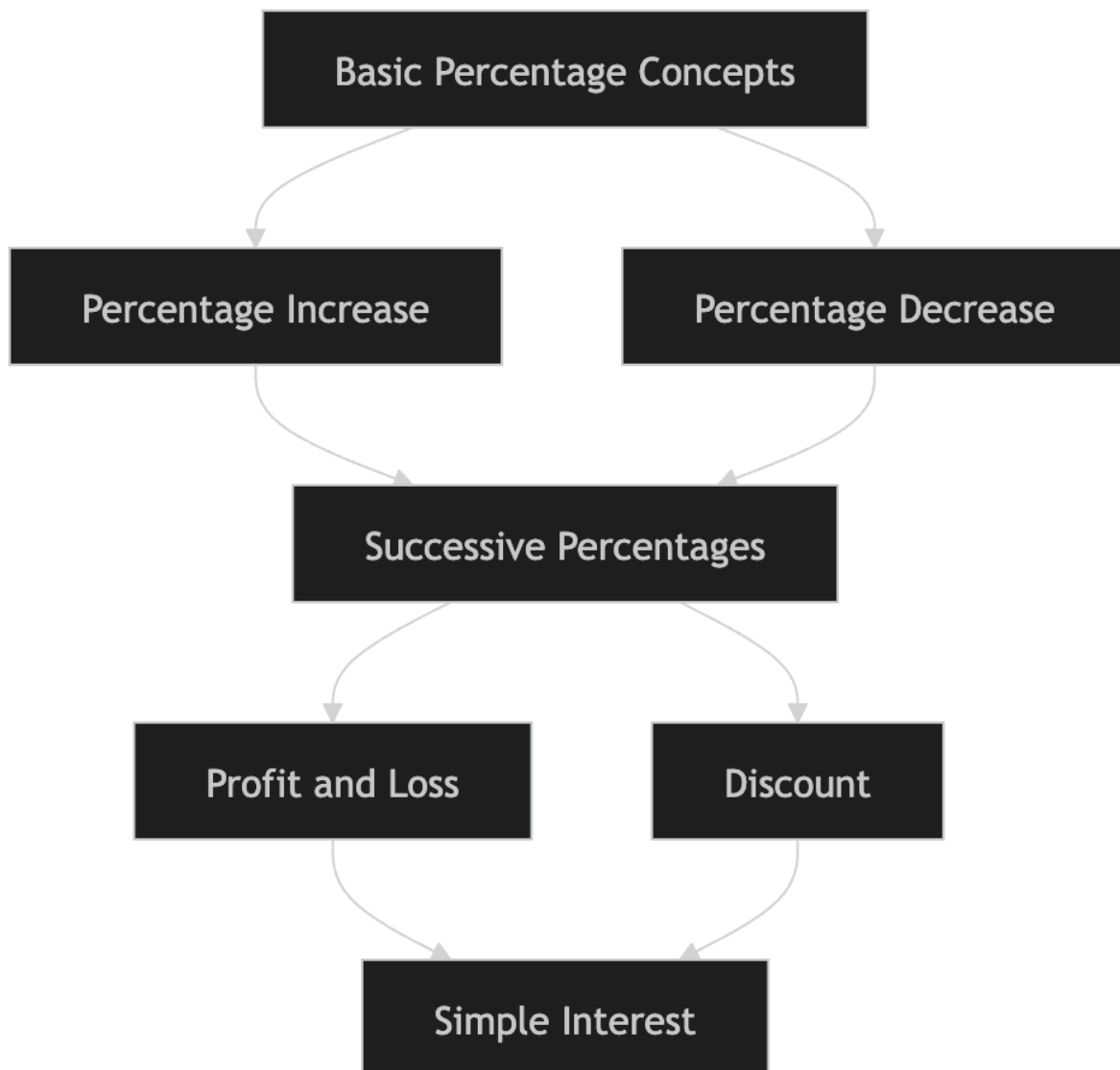
B.1 Use Case Diagram



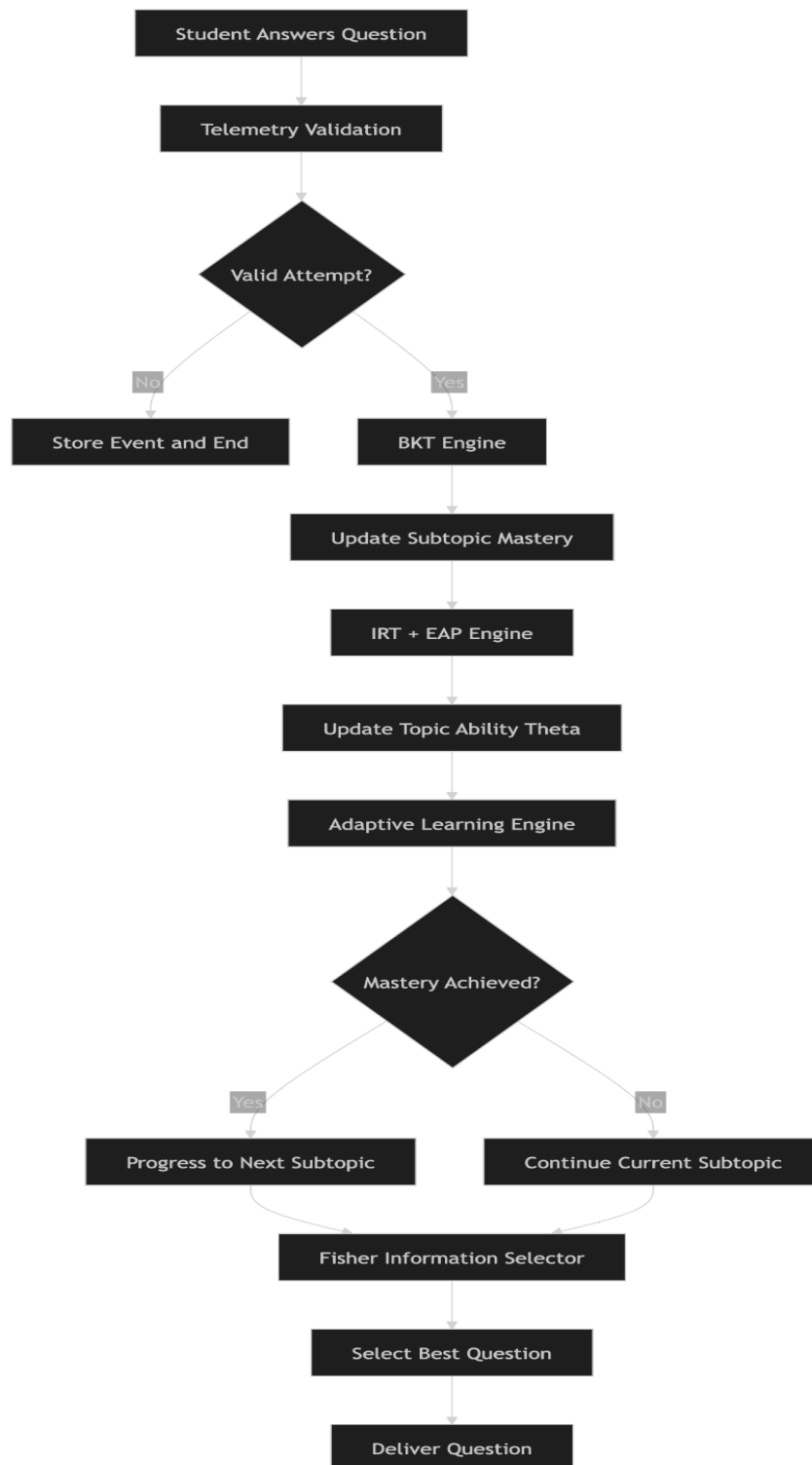
B.2 Entity Relationship Diagram (ERD)



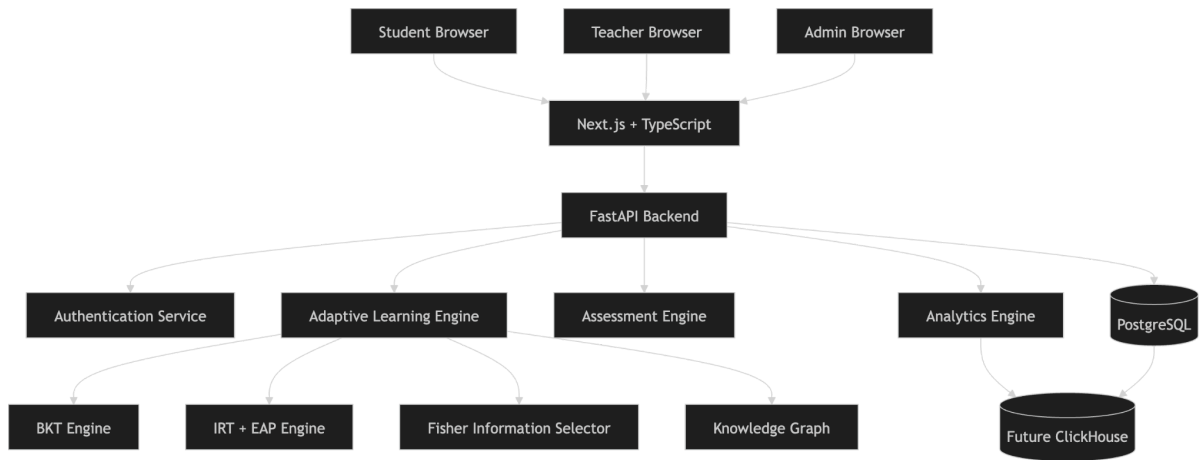
B.3 Knowledge Graph Model



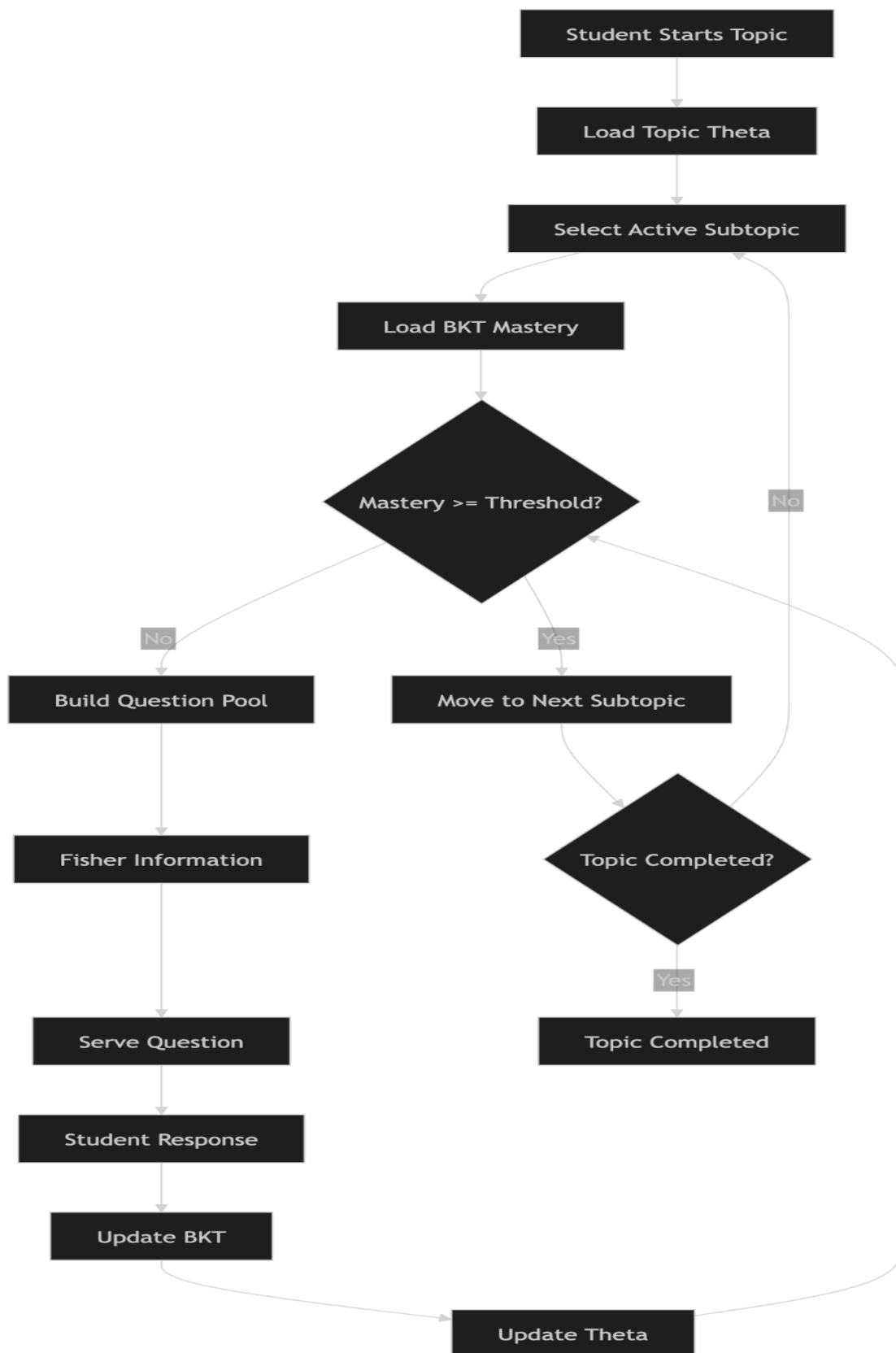
B.4 Adaptive Learning Flow Diagram



B.5 System Architecture Diagram



B.6 Core Adaptive Algorithm Architecture (Recommended Additional Diagram)



Appendix C: Issues List

The following items remain under review and may require clarification during detailed design and implementation phases.

ID	Issue	Status
ISS-001	Final BKT parameter configuration	Open
ISS-002	Final mastery threshold values	Open
ISS-003	Topic complexity tier definitions	Open
ISS-004	Initial Rasch difficulty calibration strategy	Open
ISS-005	Future 2PL migration plan	Open
ISS-006	Future MIRT adoption evaluation	Open
ISS-007	Reporting dashboard KPIs	Open
ISS-008	Notification strategy	Open
ISS-009	Infrastructure sizing estimates	Open
ISS-010	Production monitoring tools selection	Open

Appendix D: Requirements Traceability Matrix (RTM)

Requirement ID	Module	Design Phase	Development Phase	Testing Phase
FR-AUTH-*	Authentication Module	✓	✓	✓
FR-STU-*	Student Learning Module	✓	✓	✓
FR-ASM-*	Assessment Module	✓	✓	✓
FR-QBM-*	Question Bank Module	✓	✓	✓
FR-KGM-*	Knowledge Graph Module	✓	✓	✓
FR-ALE-*	Adaptive Learning Engine	✓	✓	✓
FR-BKT-*	BKT Module	✓	✓	✓
FR-IRT-*	IRT & EAP Module	✓	✓	✓
FR-FIS-*	Fisher Information Module	✓	✓	✓
FR-NAV-*	Navigation Module	✓	✓	✓
FR-ANL-*	Analytics Module	✓	✓	✓
FR-TCH-*	Teacher Portal	✓	✓	✓
FR-ADM-*	Administrator Portal	✓	✓	✓

Appendix E: Future Roadmap

Phase 1 – MVP Release

Target: Initial Production Launch

Features:

- Authentication & Authorization
- Question Bank Management
- Knowledge Graph
- BKT Engine
- Rasch IRT (1PL)
- EAP Ability Estimation
- Fisher Information Question Selection
- Adaptive Learning Engine
- Teacher Dashboard
- Administrator Dashboard
- Basic Analytics

Phase 2 – Platform Optimization

Target: 10,000+ Active Users

Features:

- 2PL IRT Migration
- Question Discrimination Calibration
- Topic-Specific BKT Parameter Optimization
- Enhanced Analytics
- Performance Dashboards
- Automated Calibration Jobs

Phase 3 – Enterprise Scale

Target: 100,000+ Active Users

Features:

- ClickHouse Analytics Platform
- Distributed Services
- Advanced Reporting
- HLR Spaced Repetition
- Large-Scale Data Processing

- Advanced Recommendation Systems

Phase 4 – AI Enhancement

Future Research Phase

Features:

- MIRT Evaluation
- Adaptive Recommendation Models
- AI Learning Assistants
- Predictive Student Risk Analysis
- Personalized Study Plans
- Intelligent Curriculum Optimization

Long-Term Vision

The ThinkPlus Adaptive Assessment Platform aims to become a scalable, psychometrically validated, AI-driven adaptive learning ecosystem capable of supporting millions of learners while maintaining explainability, reliability, and educational effectiveness.

Timeline:

Day-by-Day Schedule

Day	Task Name	Type	Dependencies	Notes / Output
1	Finalize MVP Scope, Architecture, and Delivery Plan	Doc	SRS review	Confirm MVP boundaries, role flows, major modules, acceptance criteria, and delivery assumptions. Produce initial backlog and confirm that the 22-day scope is MVP-focused.
2	Design Database Schema and ERD	Dev	Day 1	Create PostgreSQL schema for users, roles, topics, subtopics, prerequisite edges, questions, assessments, attempts, responses, mastery states, ability estimates, recommendations, and audit logs.
3	Backend Project Setup and Core Infrastructure	Dev	Day 2	Set up FastAPI project structure, configuration, database connection, migrations, Pydantic schemas, exception handling, logging, API versioning, and seed data framework.
4	Authentication and RBAC APIs	Dev	Day 3	Implement login, logout, JWT issue/validation, bcrypt password hashing, protected routes, role guards, password rules, and basic security logging.
5	User and Admin Account Management APIs	Dev	Day 4	Add admin-created users, activation/deactivation, role updates, account status, profile retrieval, failed-login tracking, and user audit trail.
6	Topic, Subtopic, and Knowledge Graph APIs	Dev	Day 3	Implement topic/subtopic CRUD, prerequisite relationships, hierarchy validation, circular dependency checks, topic search, and prerequisite retrieval.

7	Question Bank CRUD and Metadata APIs	Dev	Day 6	Build question create/edit/archive, options, answer key, explanation, topic/subtopic mapping, difficulty classification, status, and filtering APIs.
8	Question Review, Versioning, and Import Prototype	Dev	Day 7	Add draft/review/approved workflow, version history, modification audit logs, spreadsheet/CSV import validation prototype, and import summary. Risk buffer for final import format.
9	Assessment Creation and Configuration APIs	Dev	Day 7	Build assessment CRUD, type, duration, question count, topic/subtopic association, difficulty configuration, availability window, publish/status logic, and eligibility validation.
10	Assessment Delivery and Response Handling	Dev	Day 9	Implement start attempt, active state, sequential/adaptive-ready question delivery, timer data, response save/update, unanswered handling, and session continuity.
11	Assessment Evaluation and Result Storage	Dev/Test	Day 10	Evaluate responses, calculate score, topic/subtopic performance, result summaries, attempt history, and teacher review access. Milestone 1: Backend core feature review.
12	BKT Mastery Tracking Prototype	Dev	Day 10	Implement student-subtopic mastery state, BKT update after eligible response, configurable thresholds, mastery history, and signals to adaptive engine. Risk: final BKT parameters are open.
13	IRT/EAP Ability Estimation Prototype	Dev	Day 12	Implement topic-level ability estimate using MVP Rasch/IPL logic, EAP-ready service interface, ability history, confidence placeholder, and retrieval APIs. Risk:

				calibration strategy is open.
14	Fisher Information Question Selection	Dev	Day 13	Implement candidate pool filtering, exclude used questions where required, calculate information value, rank questions, select optimal question, and fallback for limited pools.
15	Adaptive Learning Engine Orchestration	Dev	Days 12-14	Connect response handling, BKT, IRT/EAP, Fisher selection, question bank, and knowledge graph to determine the next learning action/question.
16	Navigation, Backtracking, and Recommendation Logic	Dev	Day 15	Implement mastery-based progression, prerequisite-aware navigation, remediation recommendations, topic unlocking, progression history, and recommendation history. Milestone 2: Adaptive engine demo.
17	Frontend Setup, Layout, and Auth Screens	Dev	Day 4	Set up Next.js/TypeScript frontend, API client, login/logout, route protection, role-aware navigation, base dashboard layout, and error handling.
18	Admin and Content Manager Frontend Screens	Dev	Days 6-9, 17	Build screens for topic management, prerequisite management, question bank, question review/status, assessment creation, and approval workflows.
19	Student Learning and Assessment Frontend	Dev	Days 10, 15, 17	Build student dashboard, start/resume learning session, question display, answer submission, feedback, timer, assessment submission, results, and basic recommendations.
20	Teacher/Admin Analytics Dashboards	Dev	Days 11, 16, 19	Build basic dashboards for student progress, topic/subtopic mastery, assessment history, ability trends, teacher review, and admin-level summary

				metrics. Risk: final KPIs are open.
21	Integration Testing, Security Testing, and Bug Fixing	Test	Days 4-20	Run role-based flows, protected route tests, assessment attempt tests, adaptive question flow tests, import tests, browser smoke tests, API validation, and critical bug fixes. Milestone 3: UAT build.
22	Final Bug Fixes, Documentation, Deployment Handoff	Doc/Test	Day 21	Complete API documentation, deployment notes, quick user guides, known issues, final demo checklist, release notes, and sign-off package. Milestone 4: MVP delivery.

Milestone Plan

Milestone	Target Day	Acceptance Focus
Milestone 1 - Backend Core Review	Day 11	Authentication, user management, topic/question/assessment APIs, assessment delivery, evaluation, and result storage are ready for review.
Milestone 2 - Adaptive Engine Demo	Day 16	BKT, IRT/EAP prototype, Fisher selection, adaptive orchestration, navigation, and recommendation logic can demonstrate an adaptive learning loop.
Milestone 3 - UAT Build	Day 21	Integrated frontend and backend build is available for user acceptance testing with role-based flows and critical bug fixes.
Milestone 4 - MVP Delivery and Handoff	Day 22	Final demo, documentation, deployment handoff, known issues, and release notes are completed.

Risk ID	Risk / Unknown	Impact	Mitigation / Buffer Allocation
R1	Final BKT parameters are open.	High	Use configurable defaults in Day 12 prototype; document values and expose configuration for later tuning.
R2	Final mastery thresholds are open.	High	Use MVP threshold defaults and make thresholds configurable per topic/subtopic.
R3	Initial Rasch difficulty calibration strategy is open.	High	Use expert-defined difficulty mapping for MVP and maintain schema for future calibrated parameters.
R4	Reporting dashboard KPIs are open.	Medium	Build basic progress, mastery, score, attempt history, and ability trend KPIs; mark advanced KPIs for Phase 2.
R5	Notification strategy is open.	Medium	Implement in-app recommendation display only; defer email/push/WhatsApp notification channels.
R6	Infrastructure sizing and production monitoring tools are open.	Medium	Prepare deployment notes and lightweight logs; full monitoring stack remains a production hardening task.
R7	One developer timeline creates delivery pressure.	High	Keep strict MVP scope, avoid enterprise features, and protect Day 21-22 for test/fix/documentation.

